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Applied Linear Regression

Notes and Theorems



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1 Introductory Mathematical Material

1.1 Introduction

In this section we briefly introduce some familiar material from mathematics courses, namely **scalars**, **vectors** and **matrices**. We restrict our interest to vectors and matrices whose **elements** are members of the set $\mathcal{R}$ of all real numbers. We also consider some vector and matrix **operations** (transpose, addition and two kinds of multiplication), some matrix **types** (zero, square, symmetric, asymmetric, identity, singular, non-singular, orthogonal, idempotent, positive definite), some matrix **functions** (trace, rank, determinant, quadratic forms) and some matrix **constructs** (inverse, eigenroot, eigen vector).

Geometric interpretations of some concepts are supplied to assist with giving abstract concepts some realistic imagery. Some explicit test sentences are used to highlight the underlying meaning of subtle changes in equations.

The purpose of the matrix material is to create a powerful shorthand notation and background that will be useful in comprehending and handling some multivariate distributions, especially the multivariate Gaussian.

The matrix material is intended to be read and understood, as a precursor to the course, but is not directly examinable. However some matrix theory elements will need to be invoked when mastering the multivariate distribution theory, and hence will be required in examination questions on that material.

1.2 Basic concepts

A **scalar** is a number which denotes a magnitude but not a direction. We adopt the set $\mathcal{R}$ of all real numbers as the set of all scalars for our purposes, and use Greek and italicised Latin lower case symbols such as α and x , often with subscripts such as x_i , to denote such a real number.

A (column) **vector** $\mathbf{x}$ consists of n elements x_i , $i = 1, 2, \dots, n$, each of which is called a **scalar**, i.e. a real number $x_i \in \mathcal{R}$. By convention, the vector $\mathbf{x}$ is written as a column:

$$\mathbf{x}_{(n \times 1)} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

The **transpose** of $\mathbf{x}$ is a row vector denoted as $\mathbf{x}'$, and is written as

$$\mathbf{x}'_{(1 \times n)} \equiv (x_1, \dots, x_n)$$

We say the set of all possible n -dimensional vectors $\mathbf{x}$ with real number elements constitutes the vector space $\mathcal{R}^n$. The usual geometric interpretation of a vector $\mathbf{x}$ is that the elements x_i give the **co-ordinates** of the vector $\mathbf{x}$ against a set of n **orthogonal** or **perpendicular** axes within $\mathcal{R}^n$. The **origin** of those axes is designated by the position vector $\mathbf{0}$ consisting of n zeroes. Thus a vector $\mathbf{x}$ may also represent the result of a movement from the origin $\mathbf{0}$ to the position $\mathbf{x}$.

Essentially every vector $\mathbf{x}$ specifies both a direction and a distance along that direction.

We distinguish between the **dimension** n of the vector $\mathbf{x}$, and its **size** of the vector $\mathbf{x}$, which depends upon the absolute size of the elements x_i .

1.2.1 Vector sum

We define **addition** of any two vectors of the same dimension n , by addition of the corresponding elements:

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix},$$

and similarly for addition of row vectors $\mathbf{x}' + \mathbf{y}'$.

It is easily seen that vector addition is **commutative**, i.e. for any two vectors $\mathbf{x}$ and $\mathbf{y}$ vectors of the same dimension we have

$$\begin{aligned} \mathbf{x} + \mathbf{y} &= \mathbf{y} + \mathbf{x}, \\ \mathbf{x}' + \mathbf{y}' &= \mathbf{y}' + \mathbf{x}'. \end{aligned}$$

The geometric interpretation of commutativity is that a movement from one corner of a parallelogram to an opposite corner can be completed in two distinct ways, by clockwise or anti-clockwise movements along adjacent sides of the parallelogram.

Observe that there is a **zero vector** $\mathbf{0}$, all of whose elements are themselves zero, namely $0 \in \mathcal{R}$. Then

$$\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}.$$

For every vector $\mathbf{x}$ we may define a vector $-\mathbf{x}$, which is a vector of the same size as $\mathbf{x}$ but in the **opposite** direction from the geometric origin $\mathbf{0}$. Then

$$\mathbf{x} + -\mathbf{x} = -\mathbf{x} + \mathbf{x} = \mathbf{0}.$$

The geometric interpretation of this equation is that a movement in any direction $\mathbf{x}$ followed by a movement in the opposite direction of exactly the same size, brings is equivalent to no movement at all.

Also note $\mathbf{x} + \mathbf{x}$ is a column vector with each i^{th} element $2x_i$, for which we may write $\mathbf{x} + \mathbf{x} = 2\mathbf{x}$.

1.2.2 Vector products

We define **scalar multiplication** of a vector $\mathbf{x}$ (or $\mathbf{x}'$) by any scalar $\alpha \in \mathcal{R}$ as

$$\begin{aligned}\alpha.\mathbf{x}_{(n \times 1)} &= \begin{pmatrix} \alpha.x_1 \\ \vdots \\ \alpha.x_n \end{pmatrix}, \\ \alpha.\mathbf{x}'_{(1 \times n)} &\equiv (\alpha x_1, \dots, \alpha x_n).\end{aligned}$$

We use the commutativity of multiplication of real numbers ($a.b = b.a$) to write

$$\begin{aligned}\alpha.\mathbf{x} &= \mathbf{x}.\alpha, \\ \alpha.\mathbf{x}' &= \mathbf{x}'.\alpha\end{aligned}$$

In particular, setting $\alpha = 0$, we have $0.\mathbf{x} = \mathbf{0} = \mathbf{x}.0$.

By using $\alpha > 0$ the size of the vector is changed, being **shrunk** if $\alpha < 1$, and **stretched** if $\alpha > 1$. The direction is unchanged.

We permit $\alpha < 0$. These negative scalars **reverse** the direction of the vector, as well as altering the size of the vector.

The **scalar product** of two vectors $\mathbf{x}$ and $\mathbf{y}$ of the same dimension n is defined as the scalar

term

$$\begin{aligned}\mathbf{x}'\mathbf{y} &= (x_1, \dots, x_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n \\ &= \sum_{i=1}^n x_i y_i = \sum_{j=1}^n y_j x_j = \mathbf{y}'\mathbf{x}.\end{aligned}$$

The scalar product result may also be described as the product of a row vector $\mathbf{x}'$ and a column vector $\mathbf{y}$. Other names for the same operation and its result are **dot product**, or **inner product**.

Note that scalar multiplication of a vector by a scalar and scalar product of two vectors have distinct meanings.

We say two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ of the same dimension n are **orthogonal** or **perpendicular** to one another when their inner product is zero, or

$$\mathbf{x}'\mathbf{y} = \mathbf{y}'\mathbf{x} = 0.$$

If a set of k vectors $\{\mathbf{x}_i : \mathbf{x}_i \in \mathcal{R}^n, i = 1, 2, \dots, k, k \leq n\}$ is called a **mutually orthogonal set** if

$$\mathbf{x}'_i \mathbf{x}_j = 0, \text{ for } i \neq j.$$

The set of $\{\mathbf{x}_i : \mathbf{x}_i \in \mathcal{R}^n, i = 1, 2, \dots, k\}$ is called an **orthonormal set** if

$$\mathbf{x}'_i \mathbf{x}_i = 1, \text{ for all } i, \text{ and } \mathbf{x}'_i \mathbf{x}_j = 0, \text{ for } i \neq j.$$

The **maximum size** of an orthogonal set in $\mathcal{R}^n$ is $k = n$.

In contrast to the scalar product of row and column vector of the same dimensions, we may also define the **matrix product** of a column vector $\mathbf{x}$ and a row vector $\mathbf{y}'$ of arbitrary dimensions p and q respectively, as the rectangular $p \times q$ matrix

$$\begin{aligned}\mathbf{x}.\mathbf{y}' &= \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} (y_1, \dots, y_q) = \begin{pmatrix} x_1 y_1, x_1 y_2, \dots, x_1 y_q \\ x_2 y_1, x_2 y_2, \dots, x_2 y_q \\ \vdots \\ x_p y_1, x_p y_2, \dots, x_p y_q \end{pmatrix} \\ &= \begin{pmatrix} x_1 y_1 & \dots & x_1 y_q \\ \vdots & \ddots & \vdots \\ x_p y_1 & \dots & x_p y_q \end{pmatrix}.\end{aligned}$$

Similarly, we define matrix product of a column vector $\mathbf{y}$ and a row vector $\mathbf{x}'$ as the $q \times p$ matrix

$$\mathbf{y} \cdot \mathbf{x}' = \begin{pmatrix} y_1 x_1 & \dots & y_1 x_p \\ \vdots & \ddots & \vdots \\ y_q x_1 & \dots & y_q x_p \end{pmatrix}$$

Note that the dimensions of the two matrix products of vectors $\mathbf{x}$ and $\mathbf{y}$ are in general different, namely $p \times q$ and $q \times p$. In general $\mathbf{x} \cdot \mathbf{y}' \neq \mathbf{y} \cdot \mathbf{x}'$, even when the two vectors have the same dimension $p = q$.

However if in addition to $p = q$, we also have $\mathbf{x} = \mathbf{y}$, then it is easy to see that as special cases there are two fundamental **vector multiplication** operations defined on a single vector $\mathbf{x}$ of dimension p (and its transpose $\mathbf{x}'$) by

$$\mathbf{x}' \mathbf{x} = (x_1, \dots, x_p) \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} = x_1^2 + x_2^2 + \dots + x_p^2 = \sum_{i=1}^p x_i^2$$

and

$$\mathbf{x} \mathbf{x}' = \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} (x_1, \dots, x_p) = \begin{pmatrix} x_1^2, x_1 x_2, \dots, x_1 x_p \\ x_2 x_1, x_2^2, \dots, x_2 x_p \\ \vdots \\ x_p x_1, x_p x_2, \dots, x_p^2 \end{pmatrix}.$$

Thus $\mathbf{x}' \mathbf{x}$ is itself a scalar, while $\mathbf{x} \mathbf{x}'$ is a square matrix with $(p \times p)$ elements.

The scalar product $\mathbf{x}' \mathbf{x} = \sum_{i=1}^p x_i^2$ is defined to be the squared length of the vector $\mathbf{x}$.

The **norm** $\|\mathbf{x}\|$ or **length** of a vector $\mathbf{x}$ is the square root of the scalar product of $\mathbf{x}$ with itself:

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}' \mathbf{x}} = \sqrt{\sum_i x_i^2}.$$

The vector $\alpha \mathbf{x}$ has its length given by

$$\|\alpha \mathbf{x}\| = \sqrt{\alpha^2 \cdot \mathbf{x}' \mathbf{x}} = |\alpha| \cdot \|\mathbf{x}\|.$$

If a vector $\mathbf{x}$ has $\|\mathbf{x}\| = 1$, we say $\mathbf{x}$ has **unit length**. This condition is equivalent to

$$\mathbf{x}' \mathbf{x} = \|\mathbf{x}\|^2 = 1.$$

The vector $\mathbf{1} = \mathbf{1}_p = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$ is the **unit-elements vector** of dimension p and $\mathbf{1}'\mathbf{1} = 1^2 + \dots + 1^2 = p$, while

$$\mathbf{1}\mathbf{1}' = \begin{pmatrix} 1, \dots, 1 \\ 1, \dots, 1 \\ \vdots \\ 1, \dots, 1 \end{pmatrix}.$$

The norm or length of the **unit-elements vector** $\mathbf{1}$ is

$$\|\mathbf{1}\| = \sqrt{\mathbf{1}'\mathbf{1}} = \sqrt{p} = p^{\frac{1}{2}},$$

and this equation can be interpreted as the result of several applications of the Theorem of Pythagoras. Note that $\|\mathbf{1}\| > 1$, whenever $p > 1$.

1.3 Linear dependence amongst vectors

Using vector addition and scalar multiplication of vectors we define a **linear combination** of k vectors $\mathbf{x}_i$ (or $\mathbf{x}'_i$) given by k scalars α_i to be the column (or row) vector

$$\begin{aligned} \sum_{i=1}^{i=k} \alpha_i \mathbf{x}_i &= \sum_{i=1}^{i=k} \mathbf{x}_i \alpha_i, \\ \sum_{i=1}^{i=k} \alpha_i \mathbf{x}'_i &= \sum_{i=1}^{i=k} \mathbf{x}'_i \alpha_i. \end{aligned}$$

We say a set of vectors $\{\mathbf{x}_i : i = 1, 2, \dots, k\}$ are **linearly dependent** upon one another when there is *at least one* linear combination of these k vectors given by some k scalars α_i that yields $\mathbf{0}$. In that case we obtain

$$\sum_{i=1}^{i=k} \alpha_i \mathbf{x}_i = \mathbf{0}, \text{ for some scalars } \alpha_i.$$

Equivalently each n -dimensional vector $\mathbf{x}_j$ may be written as a linear combination of the other vectors in the set:

$$\mathbf{x}_j = -\alpha_j^{-1} \sum_{i \neq j} \alpha_i \mathbf{x}_i.$$

In contrast, we say a set of vectors $\{\mathbf{x}_i : i = 1, 2, \dots, k\}$ are a **linearly independent** set when there is *not even one* linear combination of these k vectors, using any choice of k scalars α_i that

yields $\mathbf{0}$, and hence

$$\sum_{i=1}^{i=k} \alpha_i \mathbf{x}_i \neq \mathbf{0}, \text{ for all possible scalars } \alpha_i.$$

The size k of a linearly independent set of n -dimensional vectors $\{\mathbf{x}_i : i = 1, 2, \dots, k\}$ must satisfy $k \leq n$.

1.4 Matrices

A **matrix** $\mathbf{A}$ is any rectangular array of elements (i.e. scalars), defined as

$$\mathbf{A} = \begin{pmatrix} a_{11}, \dots, a_{1q} \\ a_{21}, \dots, a_{2q} \\ \vdots \\ a_{p1}, \dots, a_{pq} \end{pmatrix} = (a_{ij}).$$

The matrix consists of p rows and q columns with pq cells, each containing a single scalar.

We say the **dimension** of the matrix $\mathbf{A}$ is $p \times q$, where $p > 0$ and $q > 0$, but are otherwise not restricted. Any **row vector** $\mathbf{x}$ or **column vector** $\mathbf{y}'$ can be viewed as a matrix, with either $p = 1$ or $q = 1$.

In general $\mathbf{A}$ is a rectangular matrix. For any $p \times q$ matrix $\mathbf{A}$ we define the **transpose** of $\mathbf{A}$ as the $q \times p$ matrix $\mathbf{A}'$ obtained by successively transposing each of the rows of $\mathbf{A}$ into columns (or equivalently each of the columns into rows) within $\mathbf{A}'$, so that

$$\mathbf{A}' = (a'_{ij}) = \begin{pmatrix} a_{11}, a_{21}, \dots, a_{p1} \\ a_{12}, a_{22}, \dots, a_{p2} \\ \vdots \\ a_{1q}, a_{2q}, \dots, a_{pq} \end{pmatrix} = (a_{ji}).$$

We note that performing the transpose operation twice leaves the matrix effectively unchanged.

$$\mathbf{A}'' = \mathbf{A}.$$

If $p = q$, the $p \times p$ matrix $\mathbf{A}$ is called a **square matrix**. If $\mathbf{A}$ is a square $p \times p$ matrix, then $\mathbf{A}'$ is also a square $p \times p$ matrix. In general $\mathbf{A} \neq \mathbf{A}'$, and we say the matrix $\mathbf{A}$ is **asymmetric**.

If the square matrix $\mathbf{A}$ has $a_{ij} = a_{ji}$, for all $i \neq j$, then the matrix is said to be **symmetric**. The elements of the matrix located opposite one another along the main diagonal (from top-left

to bottom-right), are equal. Equivalently, we say **A** is **symmetric** when

$$\mathbf{A} = \mathbf{A}'.$$

We may also note any **vector transpose** as a special case, so that the transpose of a row vector $\mathbf{x}'$ is a column vector $\mathbf{x}$, and conversely, and trivially

$$\mathbf{x}'' = \mathbf{x}.$$

For two $p \times q$ matrices **A** and **B** we define **matrix addition** by the addition of elements in the corresponding cells, as in the equation:

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11}, \dots, a_{1q} \\ a_{21}, \dots, a_{2q} \\ \vdots \\ a_{p1}, \dots, a_{pq} \end{pmatrix} + \begin{pmatrix} b_{11}, \dots, b_{1q} \\ b_{21}, \dots, b_{2q} \\ \vdots \\ b_{p1}, \dots, b_{pq} \end{pmatrix} = (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}).$$

The sum $\mathbf{A} + \mathbf{B}$ of the matrices **A** and **B** is the matrix of the sums $a_{ij} + b_{ij}$ of corresponding elements.

If **O** is a $p \times q$ matrix of zeroes then

$$\mathbf{A} + \mathbf{O} = \mathbf{O} + \mathbf{A} = \mathbf{A}.$$

We use **O** for matrix with all entries 0, and **0** for a vector with all its entries zero.

Note that **vector addition** of rows or of columns are special cases of matrix addition.

For a $p \times q$ matrix **A** and $q \times r$ matrix **B** we define **matrix product AB** as the $p \times r$ matrix whose $(i, j)^{th}$ element is obtained as the scalar product of the i^{th} row of **A** with the j^{th} column of **B**. Thus:

$$\mathbf{AB} = \mathbf{A} \cdot \mathbf{B} = (ab_{ij}) = (\mathbf{a}_i' \mathbf{b}_j) = \left(\sum_{k=1}^{k=q} a_{ik} b_{kj} \right).$$

The product **AB** of the matrices **A** and **B** is the matrix of the scalar products $\mathbf{a}_i' \mathbf{b}_j$ of all possible pairs of row and column vectors from **A** and **B** respectively.

However, the product **AB** can also be interpreted as the sum of q matrix products of dimension $p \times r$ obtained from the p -dimensional columns of **A** being left-multiplied on the

r -dimensional rows of $\mathbf{B}$.

$$\mathbf{AB} = \left(\sum_{k=1}^{k=q} a_{ik} b_{kj} \right) = \sum_{k=1}^{k=q} (a_{ik} b_{kj}) = \sum_{k=1}^{k=q} \mathbf{a}_k \mathbf{b}'_k.$$

The product $\mathbf{AB}$ of the matrices $\mathbf{A}$ and $\mathbf{B}$ is the sum of the matrix products $\mathbf{a}_k \mathbf{b}'_k$ of corresponding column and row vectors from $\mathbf{A}$ and $\mathbf{B}$ respectively.

In contrast to real numbers for which it is impossible to take two non-zero numbers α and β , and find $\alpha\beta = 0$, we can find **non-zero matrices** $\mathbf{A}$ and $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{O}$, regardless of whether or not $\mathbf{BA}$ exists. The product $\mathbf{AB} = \mathbf{O}$ will arise when each row vector of $\mathbf{A}$ is **orthogonal** to each column vector of $\mathbf{B}$.

Two special cases of $\mathbf{AB}$ arise: when $\mathbf{A}$ is a row vector $\mathbf{a}'$, and when $\mathbf{B}$ is a column vector $\mathbf{b}$. Then

$$\mathbf{a}'\mathbf{B} = \left(\sum_{k=1}^{k=q} a_k b_{kj} \right) = \sum_{k=1}^{k=q} (a_k b_{kj}) = \sum_{k=1}^{k=q} a_k \mathbf{b}'_k,$$

so that $\mathbf{a}'\mathbf{B}$ is a linear combination of the rows of $\mathbf{B}$. Similarly, we find

$$\mathbf{Ab} = \left(\sum_{k=1}^{k=q} a_{ik} b_k \right) = \sum_{k=1}^{k=q} (a_{ik} b_k) = \sum_{k=1}^{k=q} \mathbf{a}_k b_k,$$

and $\mathbf{Ab}$ is a linear combination of the columns of $\mathbf{A}$.

Each row $\mathbf{c}'_i$ of $\mathbf{C} = \mathbf{AB}$ is a linear combination of the rows $\mathbf{b}'_k$ of $\mathbf{B}$ given by the i^{th} row of $\mathbf{A}$:

$$\mathbf{c}'_i = \sum_{k=1}^{k=q} a_{ik} \mathbf{b}'_k.$$

Each column $\mathbf{c}_j$ of $\mathbf{C} = \mathbf{AB}$ is a linear combination of the columns $\mathbf{a}_k$ of $\mathbf{A}$, given by the j^{th} column of $\mathbf{B}$:

$$\mathbf{c}_j = \sum_{k=1}^{k=q} \mathbf{a}_k b_{kj}.$$

We say the matrices $\mathbf{A}$ and $\mathbf{B}$ are **conformable** for left-multiplication by $\mathbf{A}$ on $\mathbf{B}$ when the $p \times r$ matrix $\mathbf{AB}$ is defined, i.e. the matrices are $p \times q$ and $q \times r$ respectively. We say the matrices $\mathbf{A}$ and $\mathbf{B}$ are **conformable** for right-multiplication by $\mathbf{A}$ on $\mathbf{B}$ when the $n \times s$ matrix $\mathbf{BA}$ is defined, i.e. the matrices are $m \times s$ and $n \times m$ respectively.

In general when we can define the rectangular matrix $\mathbf{AB}$ for a $p \times q$ matrix $\mathbf{A}$ and $q \times r$ matrix $\mathbf{B}$, if $p \neq r$ there is no possible rectangular matrix $\mathbf{BA}$.

If $p = r$ then both the square $p \times p$ matrix $\mathbf{AB}$ and the square $q \times q$ matrix $\mathbf{BA}$ are defined.

In general when both $\mathbf{AB}$ and $\mathbf{BA}$ exist, they are square matrices of different dimensions unless $p = q$. Even if $p = q$, and both left-product and right-product of the two $p \times p$ matrices $\mathbf{A}$ and $\mathbf{B}$ have the same dimension $p \times p$ in general we have

$$\mathbf{AB} \neq \mathbf{BA}.$$

Thus matrix multiplication of arbitrary but conformable square matrices is in general **non-commutative**. However it is possible that the matrices $\mathbf{AB}$ and $\mathbf{BA}$ may be equal when some particular conditions hold for $\mathbf{A}$ and $\mathbf{B}$. Under those conditions only we find $\mathbf{AB} = \mathbf{BA}$, as a special case.

The **transpose** of a matrix product is the **reversed product** of the matrix transposes:

$$(\mathbf{AB})' = \left(\sum_{k=1}^{k=q} a_{ik} b_{kj} \right)' = \left(\sum_{k=1}^{k=q} a_{jk} b_{ki} \right) = \left(\sum_{k=1}^{k=q} a'_{kj} b'_{ik} \right) = \left(\sum_{k=1}^{k=q} b'_{ik} a'_{kj} \right) = \mathbf{B}' \mathbf{A}'.$$

1.4.1 Operations on matrices

We say we perform **elementary row operations** on the matrix $\mathbf{A}$ when we interchange any pair of rows, or multiply a particular row by a scalar, or add a multiple of one row to another row. Every elementary row operation on $\mathbf{A}$ can be represented by left-multiplication or pre-multiplication of $\mathbf{A}$ by a suitable square matrix, say $\mathbf{R}_i$, of one of the following types:

$$\mathbf{R}_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{R}_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{R}_3 = \begin{pmatrix} 1 & \alpha & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

These matrices respectively switch rows three and four, multiply row 2 by a scalar and add a scalar multiple of row 2 to row one.

Similarly, we may describe elementary column operations on the matrix $\mathbf{A}$ by post-multiplying the matrix $\mathbf{A}$ by corresponding matrices $\mathbf{C}_i$.

1.4.2 Trace

The **trace** $tr(\mathbf{A})$ of a square $p \times p$ matrix $\mathbf{A}$ is the sum of its diagonal elements:

$$tr(\mathbf{A}) = \sum_{i=1}^{i=p} a_{ii}.$$

Providing that both square matrix products $\mathbf{AB}$ and $\mathbf{BA}$ exist, even if they have different dimensions $p \times p$ and $q \times q$, we have

$$tr(\mathbf{AB}) = \sum_{i=1}^{i=p} \sum_{j=1}^{j=q} a_{ij} b_{ji} = \sum_{j=1}^{j=q} \sum_{i=1}^{i=p} b_{ji} a_{ij} = tr(\mathbf{BA}).$$

A consequence of this result is that for arbitrary conformable matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$, we obtain

$$tr(\mathbf{ABC}) = tr(\mathbf{BCA}) = tr(\mathbf{CAB}),$$

and, as a special case for arbitrary conformable vectors $\mathbf{x}$ and $\mathbf{y}$, of possibly different dimensions, we have

$$tr(\mathbf{x}'\mathbf{A}\mathbf{y}) = tr(\mathbf{A}\mathbf{y}\mathbf{x}') = tr(\mathbf{y}\mathbf{x}'\mathbf{A}).$$

If matrix $\mathbf{A}$ is rectangular we may consider the case when $\mathbf{B} = \mathbf{A}'$. Then both matrix products $\mathbf{AB}$ and $\mathbf{BA}$ exist, and in fact $\mathbf{AB} = \mathbf{AA}'$ and $\mathbf{BA} = \mathbf{A}'\mathbf{A}$. Both $\mathbf{AA}'$ and $\mathbf{A}'\mathbf{A}$ are **square** matrices, both are **symmetric**, but in general because their dimensions are different, we have

$$\mathbf{AA}' \neq \mathbf{A}'\mathbf{A}.$$

Hence, even though $\mathbf{AA}' \neq \mathbf{A}'\mathbf{A}$, we always have

$$tr(\mathbf{AA}') = \sum_{i=1}^{i=p} \sum_{j=1}^{j=q} a_{ij} a_{ji} = \sum_{j=1}^{j=q} \sum_{i=1}^{i=p} a_{ji} a_{ij} = tr(\mathbf{A}'\mathbf{A}).$$

In particular, for $\mathbf{A} = \mathbf{x}$ we have

$$tr(\mathbf{xx}') = \sum_{i=1}^{i=p} x_{ii}^2 = tr(\mathbf{x}'\mathbf{x}).$$

If matrix $\mathbf{A}$ is square $p \times p$, then $\mathbf{AB} = \mathbf{AA}'$ and $\mathbf{BA} = \mathbf{A}'\mathbf{A}$ are different square matrices of the

same dimension $p \times p$. Thus in general

$$\mathbf{A}\mathbf{A}' \neq \mathbf{A}'\mathbf{A},$$

but an equality may hold in some circumstances, e.g. when $\mathbf{A}$ is symmetric.

1.4.3 Rank

The **rank** $r(\mathbf{A})$ of any $p \times q$ matrix $\mathbf{A}$ is the largest possible number k of linearly independent rows in the matrix. This number k is equal to the largest possible number of linearly independent columns within the matrix $\mathbf{A}$. Note that $r(\mathbf{A}) = k \leq \min(p, q)$. If $k = p$, we say the matrix $\mathbf{A}$ has **full row rank**, and if $k = q$, we say $\mathbf{A}$ has **full column rank**. By considering linear combinations of rows and columns we can establish the identity

$$r(\mathbf{A}) = r(\mathbf{A}') = r(\mathbf{A}'\mathbf{A}) = r(\mathbf{A}\mathbf{A}').$$

For the zero matrix $\mathbf{O}$ we define $r(\mathbf{O}) = 0$, and for vectors $\mathbf{x}$ we have $r(\mathbf{x}) = 1$.

It follows from the definitions of rank and of matrix multiplication that

$$r(\mathbf{AB}) \leq \min\{r(\mathbf{A}), r(\mathbf{B})\}.$$

When both matrix products $\mathbf{AB}$ and $\mathbf{BA}$ exist, in general $r(\mathbf{AB}) \neq r(\mathbf{BA})$, but rank equality may hold under additional conditions.

1.4.4 Identity, idempotence and rank

Any square $n \times n$ matrix $\mathbf{I}_n$ with each of its n diagonal elements equal to 1, and all its off-diagonal elements equal to 0, is an **identity matrix**. Any identity matrix is **symmetric**. For an arbitrary $p \times q$ matrix $\mathbf{A}$ we have

$$\mathbf{I}_p \cdot \mathbf{A} = \mathbf{A} = \mathbf{A} \cdot \mathbf{I}_q,$$

and for every square $p \times p$ matrix $\mathbf{A}$ we have $\mathbf{I}_p \cdot \mathbf{A} = \mathbf{A} = \mathbf{A} \cdot \mathbf{I}_p$.

We note that the rows of every identity matrix $\mathbf{I}_p$ all have **unit length**, and are also **mutually orthogonal** row vectors. Similarly, the columns of $\mathbf{I}_p$ all have unit length and are mutually orthogonal column vectors. It is also clear that the rows (and the columns) of $\mathbf{I}_p$ are **linearly independent** of one another. Note that $\mathbf{I}_p$ has $p = r(\mathbf{I}_p) = \text{tr}(\mathbf{I}_p)$.

Every identity matrix is **idempotent**, because $\mathbf{I}_p^2 = \mathbf{I}_p \cdot \mathbf{I}_p = \mathbf{I}_p$, and hence all k^{th} powers of $\mathbf{I}_p$

reduce to $\mathbf{I}_p$ itself:

$$\mathbf{I}_p^k = \mathbf{I}_p \cdot \mathbf{I}_p \dots \mathbf{I}_p = \mathbf{I}_p, \text{ for every power } k.$$

The identity matrix $\mathbf{I}_p$ is unique for a p -dimensional space, but many non-identity $p \times p$ matrices also share the property of idempotence.

For instance, the square $p \times p$ zero matrices $\mathbf{O}_p$ are idempotent with

$$\mathbf{O}_p^k = \mathbf{O}_p \cdot \mathbf{O}_p \dots \mathbf{O}_p = \mathbf{O}_p, \text{ for every power } k.$$

Again square $p \times p$ matrices $\mathbf{P}$ with structures of the form

$$\mathbf{P} = \begin{pmatrix} \mathbf{I}_r & \mathbf{K} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}, \quad \text{or} \quad \mathbf{P} = \begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{L} & \mathbf{O} \end{pmatrix}.$$

where $\mathbf{K}$ is an arbitrary $r \times (p - r)$ matrix, and $\mathbf{L}$ is an arbitrary $(p - r) \times r$ matrix, all satisfy the condition $\mathbf{P}^2 = \mathbf{P}$, and hence are idempotent. These examples show idempotent matrices can be **symmetric** or **asymmetric**. Observe that for idempotent matrices $\mathbf{P}$ we have

$$k = r(\mathbf{P}) = r(\mathbf{P}') = r(\mathbf{P}'\mathbf{P}) = r(\mathbf{P}\mathbf{P}') = tr(\mathbf{P}) = tr(\mathbf{P}') = tr(\mathbf{P}'\mathbf{P}) = tr(\mathbf{P}\mathbf{P}').$$

If $\mathbf{P}$ is a $p \times p$ idempotent matrix, then $(\mathbf{I}_p - \mathbf{P})$ is also idempotent. The proof uses the equation $\mathbf{P}(\mathbf{I}_p - \mathbf{P}) = \mathbf{O}$. We will find many other idempotent matrices in multivariate statistics theory. They will also have the property that rank and trace are equal.

1.4.5 Inverses

For some square $p \times p$ matrices $\mathbf{A}$ there exists a unique (multiplicative) **inverse matrix** of $\mathbf{A}$, denoted as $\mathbf{A}^{-1}$, with the property that

$$\mathbf{A}^{-1} \cdot \mathbf{A} = \mathbf{I}_p = \mathbf{A} \cdot \mathbf{A}^{-1}.$$

If the inverse $\mathbf{A}^{-1}$ does exist we say $\mathbf{A}$ is a **non-singular** matrix, and that $\mathbf{A}$ is **invertible**.

If the inverse does not exist we say $\mathbf{A}$ is a **singular** matrix, and is **non-invertible**.

The $p \times p$ matrix inverse $\mathbf{A}^{-1}$ will only exist when the square $p \times p$ matrix $\mathbf{A}$ has full row and column rank $k = r(\mathbf{A}) = p$. In that case we can find $\mathbf{A}^{-1}$ by performing **elementary row operations** on the rectangular $p \times 2p$ matrix $\begin{pmatrix} \mathbf{A} & \mathbf{I}_p \end{pmatrix}$, until we obtain a rectangular matrix $\begin{pmatrix} \mathbf{I}_p & \mathbf{B} \end{pmatrix}$. Then the square matrix $\mathbf{B}$ is the inverse of $\mathbf{A}$, and we write $\mathbf{B} = \mathbf{A}^{-1}$.

In contrast, if the inverse of a square matrix $\mathbf{A}$ does not exist, and $\mathbf{A}$ has rank $r = r(\mathbf{A}) < p$,

by elementary row operations we will obtain a matrix of a different form, namely $\begin{pmatrix} \mathbf{K}_p & \mathbf{B} \end{pmatrix}$, where $\mathbf{K}_p$ is of the type

$$\mathbf{K}_p = \begin{pmatrix} \mathbf{I}_r & \mathbf{L} \\ \mathbf{O} & \mathbf{O}_{p-r} \end{pmatrix}.$$

Note the contrast between $\begin{pmatrix} \mathbf{I}_p & \mathbf{B} \end{pmatrix}$ and $\begin{pmatrix} \mathbf{K}_p & \mathbf{B} \end{pmatrix}$.

Of the three types of matrices $\mathbf{R}_i$ for **elementary row operations**, all three are invertible, only $\mathbf{R}_1$ is its own inverse, and none are idempotent.

1.4.6 Orthonormal matrices

If a square $p \times p$ matrix $\mathbf{H}$ has its **inverse** equal to its **transpose** then we have

$$\mathbf{H}^{-1} = \mathbf{H}' \quad \text{and} \quad \mathbf{H}' \cdot \mathbf{H} = \mathbf{I}_p = \mathbf{H} \cdot \mathbf{H}'$$

Then we say such an $\mathbf{H}$ is an **orthogonal matrix** or more strictly an **orthonormal matrix**. Note that $\mathbf{I}_p$ is an orthogonal matrix.

The **matrix product** of any two orthogonal matrices is itself **orthogonal**, since

$$(\mathbf{HJ})' = \mathbf{J}'\mathbf{H}' \quad \text{and} \quad \mathbf{HJ} \cdot \mathbf{J}'\mathbf{H}' = \mathbf{H}\mathbf{I}_p\mathbf{H}' = \mathbf{H}\mathbf{H}' = \mathbf{I}_p.$$

Of the matrices for **elementary row operations**, only $\mathbf{R}_1$ is orthogonal.

1.4.7 Determinants

The **determinant** of a square $p \times p$ matrix $\mathbf{A}$ is a scalar designated by the notation $\det(\mathbf{A})$ or $|\mathbf{A}|$. The definition is complicated:

$$\det(\mathbf{A}) = \sum_{\theta} \text{sgn}(\theta) \prod_{i=1}^{i=p} a_{i\theta(i)} = |\mathbf{A}|$$

where θ varies in turn through all $p!$ permutations on the numbers $1, 2, \dots, p$, and $\text{sgn}(\theta) = (-1)^k$, where k is the number of pairwise switches necessary to obtain the order for permutation θ from the ordered set $1, 2, \dots, p$.

The quantity $\det(\mathbf{A}) = |\mathbf{A}|$ is permitted to be **negative**, so that the notation should be distinguished from the **modulus** or absolute value of a scalar, for which $\text{mod}(a) = |\alpha| = \alpha$ for $\alpha > 0$, and $\text{mod}(a) = |\alpha| = -\alpha$ for $\alpha < 0$.

We can show that if $r(\mathbf{A}) < p$, then $\det(\mathbf{A}) = 0$,. and if $r(\mathbf{A}) = p$, then $\det(\mathbf{A}) \neq 0$.

There are however many ways to obtain the determinant by successive **iterations**. For a square 2×2 matrix $\mathbf{A}$, we have $p! = 2! = 2$ terms in the summation:

$$\det(\mathbf{A}) = a_{11}a_{22} - a_{21}a_{12} = a_{11}(a_{22} - a_{21}a_{11}^{-1}a_{12}) = a_{22}(a_{11} - a_{12}a_{22}^{-1}a_{21}).$$

For a square 3×3 matrix $\mathbf{A}$, we have $p! = 3! = 6$ terms in the summation:

$$\begin{aligned} \det(\mathbf{A}) &= a_{11}a_{22}a_{33} - a_{12}a_{21}a_{33} + a_{13}a_{23}a_{32} - a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} - a_{13}a_{22}a_{31}, \\ &= a_{33} \begin{vmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}. \end{aligned}$$

This pattern allows us to find any suitable set of arbitrary partitions of the $p \times p$ matrix $\mathbf{A}$ with **square submatrices** on the major diagonal (top-left to bottom-right), namely

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix},$$

all of which yield lead to the unique value of the determinant $|\mathbf{A}|$ through the many formulae of the type

$$|\mathbf{A}| = |\mathbf{A}_{11}| |\mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}| = |\mathbf{A}_{22}| |\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}|.$$

being applied iteratively.

If either $\mathbf{A}_{21} = \mathbf{O}$ or $\mathbf{A}_{12} = \mathbf{O}$, then

$$|\mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}| = |\mathbf{A}_{22}| \quad \text{and} \quad |\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}| = |\mathbf{A}_{11}|.$$

For the determinant of the product of square matrices $\mathbf{A}$ and $\mathbf{B}$ we obtain the product of the determinants:

$$|\mathbf{AB}| = |\mathbf{A}| |\mathbf{B}| = |\mathbf{B}| |\mathbf{A}| = |\mathbf{BA}|.$$

Hence for inverse matrices we have the determinant of the inverse of $\mathbf{A}$ is the inverse of the determinant of $\mathbf{A}$

$$|\mathbf{A}^{-1}| = |\mathbf{A}|^{-1}.$$

For orthogonal matrices $\mathbf{H}$ we have

$$|\mathbf{H}'| = |\mathbf{H}^{-1}| = |\mathbf{H}| = 1.$$

The geometric interpretation of the determinant is the **hypervolume** of the **hyperparelloiped** based at the origin and with **edges** given by the **rows** of the matrix. Equivalently it is the hypervolume of the hyperparelloiped based at the origin and with edges given by the **columns** of the matrix.

1.4.8 Eigenstructure

In general for any rectangular matrix **A** we may find some pairs of scalars λ and vectors **x** such that .

$$\mathbf{Ax} = \lambda\mathbf{x}, \text{ or } (\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}.$$

The vector **x** is called a (column) **eigenvector** or **characteristic** vector or **latent vector** of **A** corresponding to **eigenvalue** or **characteristic root** or **latent root** λ . Similarly one may find row eigenvectors and eigen roots. The eigenvalues for rows and for columns are identical, being the solutions λ_i for which

$$|\mathbf{A} - \lambda\mathbf{I}| = |\mathbf{A}' - \lambda\mathbf{I}| = 0.$$

We note without proof here that for any $p \times p$ symmetric matrix **A** there always exists a $p \times p$ **orthogonal matrix P** such that

$$\mathbf{P}'\mathbf{AP} = \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$$

where the λ_i are the eigenvalues of **A**.

The row and column eigenvectors of a **symmetric** matrix are **equivalent**.

If all the eigenvalues or characteristic roots λ_i of **A** satisfy $\lambda_i > 0$ we say **A** is **positive definite**. Then

$$|\mathbf{A}| = |\mathbf{\Lambda}| = \prod_{i=1}^{i=p} \lambda_i > 0.$$

If **A** is of rank $k \leq p$ and if the roots λ_i (and vectors in **P**) are ordered from large to small then

$$\lambda_i = 0, \text{ for } i = k + 1 \text{ to } p.$$

We will assume **A** is positive definite, that is of **full rank** p . Thus we can compute on-zero values $\lambda^{1/2}$ and $\lambda^{-1/2}$.

Let

$$\mathbf{\Lambda}^{-1/2} = \text{diag}(\lambda_1^{-1/2}, \dots, \lambda_p^{-1/2})$$

Then

$$\Lambda^{-1/2} \Lambda \Lambda^{-1/2} = \mathbf{I}$$

Now

$$\Lambda^{-1/2} \mathbf{P}' \mathbf{A} \mathbf{P} \Lambda^{-1/2} = \Lambda^{-1/2} \Lambda \Lambda^{-1/2} = \mathbf{I}.$$

Let

$$\mathbf{C} = \mathbf{P} \Lambda^{-1/2}, \quad \mathbf{C}' = \Lambda^{-1/2} \mathbf{P}'$$

then $\mathbf{C}' \mathbf{A} \mathbf{C} = \mathbf{I}$ and $\mathbf{C}^{-1}$ exists, that is $\mathbf{C}$ is a **non-singular** matrix. Thus for $\mathbf{A}$ symmetric and invertible with positive eigenroots there will always exist an invertible matrix $\mathbf{C}$ such that

$$\mathbf{C}' \mathbf{A} \mathbf{C} = \mathbf{I}_p.$$

The **eigenvalues** of an **idempotent** matrix $\mathbf{A}$ are the solutions to

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{x} = \mathbf{0}, \quad \mathbf{A} \mathbf{x} = \lambda \mathbf{x},$$

$$\mathbf{A}^2 \mathbf{x} = \lambda \mathbf{A} \mathbf{x} = \lambda^2 \mathbf{x}, \text{ after premultiplying by } \mathbf{A}.$$

Thus

$$\mathbf{A}^2 \mathbf{x} = \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$$

hence $\lambda \mathbf{x} = \lambda^2 \mathbf{x}$, $(\lambda - \lambda^2) \mathbf{x} = \mathbf{0}$, thus $\lambda = 0$ or $\lambda = 1$. The eigenvalues of a matrix $\mathbf{A}$ are all either 0's or 1's.

Where $\mathbf{A}$ is **symmetric** and **idempotent** of rank k , there exists an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}' \mathbf{A} \mathbf{P} = \begin{pmatrix} \mathbf{I}_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$

For any symmetric matrix $\mathbf{A}$ there exists an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}' \mathbf{A} \mathbf{P} = \Lambda$, $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$ where the λ_i are the eigenvalues of $\mathbf{A}$. But here $\mathbf{A}$ is of rank k , so there are only k such eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$, the rest of the eigenvalues must be zero. Since $\mathbf{A}$ is idempotent, all the λ_i must equal 1 for $i = 1, \dots, k$. We can always arrange the columns of $\mathbf{P}$ such that $\mathbf{P}' \mathbf{A} \mathbf{P} = \begin{pmatrix} \mathbf{I}_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$.

If the $n \times q$ matrix $\mathbf{X}$ has full column rank q , then $\mathbf{X}' \mathbf{X}$ is $q \times q$ matrix of rank q , and has an inverse $(\mathbf{X}' \mathbf{X})^{-1}$. The $n \times n$ matrix $\mathbf{C} = \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$ has rank q , and does not have an inverse. However $\mathbf{C}$ is **idempotent**, because $\mathbf{C}^2 = \mathbf{C}$, with $\text{tr}(\mathbf{C}) = q$.

We note that $(\mathbf{I}_n - \mathbf{C}) = (\mathbf{I}_n - \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}')$ is also idempotent with $\text{tr}(\mathbf{I}_n - \mathbf{C}) = n - \text{tr}(\mathbf{C}) = n - q$.

1.4.9 Quadratic forms

If $\mathbf{A}$ is **symmetric** and $(p \times p)$ then we define a **quadratic form** in $\mathbf{A}$ as the product

$$\begin{aligned}\mathbf{x}'\mathbf{A}\mathbf{x} &= (x_1, \dots, x_p) \begin{pmatrix} a_{11}, \dots, a_{1p} \\ \vdots \\ a_{p1}, \dots, a_{pp} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} \\ &= a_{11}x_1^2 + 2a_{12}x_1x_2 + \dots + 2a_{1p}x_1x_p + a_{22}x_2^2 + 2a_{2p}x_2x_p + \dots + a_{pp}x_p^2 \\ &= \sum_i \sum_j a_{ij}x_i x_j \text{ with } a_{ij} = a_{ji} \text{ } i \neq j.\end{aligned}$$

We say the quadratic form is **non-negative definite** when $\mathbf{x}'\mathbf{A}\mathbf{x} \geq 0$, for all non-zero $\mathbf{x}$. The form is **positive definite** when $\mathbf{x}'\mathbf{A}\mathbf{x} > 0$, for all non-zero $\mathbf{x}$. This condition holds whenever $\mathbf{A}$ can be written as $\mathbf{B}\mathbf{B}'$ (or $\mathbf{B}'\mathbf{B}$) for some rectangular matrix $\mathbf{B}$. For simplicity we will assume that all quadratic forms we discuss are positive definite. This approach amounts to assuming all the **symmetric** matrices we use will have **positive eigenvalues**.

A geometric interpretation of a non-negative definite quadratic form arises from setting $\mathbf{x}'\mathbf{A}\mathbf{x} = k > 0$. As k increases from zero, we obtain a monotonic family of the outer surfaces of larger and larger **ellipsoids** in p -dimensional space, all centred upon the origin $\mathbf{0}$.

If $\mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_p \end{pmatrix}$, then $\mathbf{x} - \mathbf{b} = \begin{pmatrix} x_1 - b_1 \\ \vdots \\ x_p - b_p \end{pmatrix}$. Then similarly, for $\mathbf{x} - \mathbf{b}$ we have another quadratic form

$$\begin{aligned}(\mathbf{x} - \mathbf{b})'\mathbf{A}(\mathbf{x} - \mathbf{b}) &= \sum_i \sum_j a_{ij}(x_i - b_i)(x_j - b_j) \\ &= \mathbf{x}'\mathbf{A}\mathbf{x} - 2\mathbf{x}'\mathbf{A}\mathbf{b} + \mathbf{b}'\mathbf{A}\mathbf{b}.\end{aligned}$$

Then by increasing k in the equations $(\mathbf{x} - \mathbf{b})'\mathbf{A}(\mathbf{x} - \mathbf{b}) = k$ we obtain another monotonic family of the outer surfaces of ever-increasing hyperellipsoids in p -dimensional space, all centred at the point with co-ordinates $\mathbf{b}$. When $\mathbf{A} = \mathbf{I}_p$ the hyperellipsoids are in fact **hyperspheres** in p -dimensional space, centred at $\mathbf{0}$ or at $\mathbf{b}$. The **interior** regions of the hyperellipsoids may be designated by $\mathbf{x}'\mathbf{A}\mathbf{x} < k$.

The **hypervolume** of hyperspace within a **hyperellipsoid** is related to the hypervolume of the rectangular **hyperblock** which encloses it completely, in much the same way that the area of a circle is related to the area of the smallest square that encloses it, or the volume of a sphere is related to the volume of the smallest cube that encloses the sphere.

In the statistics course we will be interested in hypervolumes of **integrand** functions other than the uniform value 1 applicable to area, volume and hypervolume calculations. In fact

we will give attention to **probability density functions** which have their maximum values at the centre of the hyperellipsoid, and for which the density values diminish rapidly with distance from the centre.

By using **positive definite quadratic forms** we will ensure that the diminishing density values define **hypercontours** in hyperspace corresponding to the hyperellipsoid surfaces on each of which the density function assumes a common value. The multivariate Gaussian distribution will be a main focus of the course. The density involves a term of the form

$$\exp\left[-\frac{1}{2}\mathbf{x}'\mathbf{A}\mathbf{x}\right] \text{ with contours given by } \mathbf{x}'\mathbf{A}\mathbf{x} = k.$$

2 Some Theory about the Multivariate Normal Distribution

All proofs for theorems are provided in chapter 12 from pages 80.

Proposition 1 *The density of the multivariate normal distribution of a random vector $\mathbf{X}$, is given by*

$$f(x_1, \dots, x_p) = \frac{1}{(2\pi)^{\frac{1}{2}p} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\mu)' \Sigma^{-1}(\mathbf{x}-\mu)}$$

with $E(\mathbf{X}) = \mu$, $\text{cov}(\mathbf{X}) = E((\mathbf{X} - \mu)(\mathbf{X} - \mu)') = \Sigma$.

Theorem 2 *If $\mathbf{X} \sim N(\mu, \Sigma)$, then the moment generating function of $\mathbf{X}$ is:*

$$M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}}) = e^{\mathbf{t}'\mu + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t}}$$

Theorem 3 *If $\mathbf{X} \sim N(\mu, \Sigma)$, then if $\mathbf{Y} = \mathbf{C}\mathbf{X}$ for any matrix/vector $\mathbf{C}$ then $\mathbf{Y} \sim N(\mathbf{C}\mu, \mathbf{C}\Sigma\mathbf{C}')$.*

Theorem 4 *Let $\mathbf{X} \sim N(\mu, \Sigma)$ and let $\mathbf{X}$ be partitioned into $\mathbf{X} = \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix}$, then $\mathbf{X}_{(q \times 1)}^{(1)} \sim N(\mu^{(1)}, \Sigma_{11})$ and $\mathbf{X}_{(r \times 1)}^{(2)} \sim N(\mu^{(2)}, \Sigma_{22})$ where*

$$\mu = \begin{pmatrix} \mu^{(1)} \\ \mu^{(2)} \end{pmatrix}_r^q$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{21} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}_r^q$$

Theorem 5 *If $\mathbf{Y}_{(n \times 1)} \sim N(\mathbf{O}, \mathbf{I}_n)$, then $\mathbf{Y}'\mathbf{A}\mathbf{Y}$ is distributed as χ_k^2 , where $\mathbf{A}$ is idempotent of rank k .*

Theorem 6 *If $\mathbf{Y}_{(n \times 1)} \sim N(\mathbf{O}, \mathbf{I}_n)$, then the linear form $\mathbf{B}\mathbf{Y}$ is independent of the quadratic form $\mathbf{Y}'\mathbf{A}\mathbf{Y}$ ($\mathbf{A}$ idempotent of rank k) if $\mathbf{B}\mathbf{A} = \mathbf{O}$.*

3 The General Linear Model

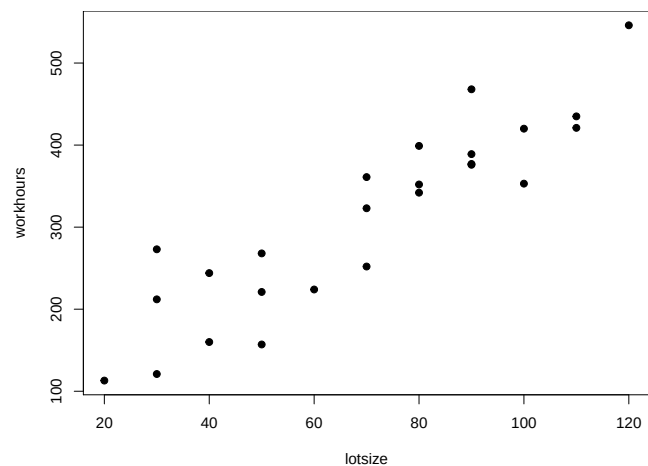
3.1 Introductory Example

The following example is taken from Neter et al (1990). The Toluca Company produces refrigeration equipment. One of its parts is produced in lots of varying sizes. In order to streamline costs management suggested that the company identify the optimum lot size for production. In order to tackle this problem an analyst suggested that the optimal lot size is linked to the labour hours needed to produce various lots. Figure 1 below displays the lot size (X) and the number of hours needed (work hours (Y)) to produce the various lots. The relationship between the X (the explanatory variable) and Y (the response variable) appears to be reasonably linear. The analyst collected data for 25 lot sizes (n). We will represent the i^{th} observation pair as (X_i, Y_i) . e.g. $X_9 = 100$ and $Y_9 = 353$.

```
toluca <- read.table("toluca.txt",
                     header = T)
head(toluca, n = 9)
```

##	lotsize	workhours
## 1	80	399
## 2	30	121
## 3	50	221
## 4	90	376
## 5	70	361
## 6	60	224
## 7	120	546
## 8	80	352
## 9	100	353

```
attach(toluca)
par(mar = c(4,4,0,0))
plot(lotsize, workhours, pch = 19)
```



In this course we will investigate how to estimate this linear relationship. We will also consider introducing more than one explanatory variable and identify which of these variables might be more important (variable selection). e.g. the production costs of the different lots sizes might be different due and this might influence the optimal lot size of production. In such a case the data is stored in a matrix $\mathbf{X} = (X_1, \dots, X_p)$ such that each column of $\mathbf{X}$ represents a different variable. The elements are referenced as X_{ij} . i indicates the row number while j represents the column number. On occasion we might have outliers (too large or small observations) which requires special techniques when fitting regression models. Some other topics considered are model selection, handling indicator variables and transformations.

3.2 The General Linear Model

We now generalise the above setting by considering more than one explanatory variable. Consider the situation in which a random variable $\mathbf{Y}$, called the response variable depends (read this as *is a function of*) on p explanatory variables $X_1, \dots, X_p$. We assume that we can model Y as a linear function of the X variables. It is also assumed that the relationship might not be perfect such that

$$\mathbf{Y} = \beta_0 + \beta_1 \mathbf{X}_1 + \dots + \beta_p \mathbf{X}_p + \mathbf{e} \quad (1)$$

We further assume that the residual vector ($\mathbf{e}$) is a random variable such that $E(\mathbf{e}) = \mathbf{0}$ and $E(\mathbf{e}\mathbf{e}') = \sigma^2 \mathbf{I}_n$ where n is the number of observations in the data set. $E(\mathbf{e}\mathbf{e}')$ is the expected covariance matrix of the residual vector.

The β 's and σ^2 are estimated by taking a random sample from $\mathbf{Y}$ and the corresponding observations from the $\mathbf{X}$ variables. Denote these observations by

$$(Y_i, X_{i1}, \dots, X_{ip}), i = 1, \dots, n$$

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & X_{12} & \dots & X_{1p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{np} \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \quad \mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}$$

Note that $\mathbf{Y}$ is $(n \times 1)$, $\mathbf{X}$ is $(n \times k)$, β is $(k \times 1)$ and $\mathbf{e}$ is $(n \times 1)$, where $k = p + 1$ and $n \gg p$. Equation (1) can be rewritten as

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{e} \quad (2)$$

which is called **The General Linear Model**.

3.3 Maximum Likelihood Estimates and Distributions

Let $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ with $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ (read as *e is distributed as a multivariate normal random variable with a zero mean vector and a diagonal variance covariance matrix.*) If $(e_1, \dots, e_n)$ is a random

sample from $N(0, \sigma^2)$ then from estimation theory the likelihood equation is given by

$$L(\beta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}n}} \exp\left(-\frac{(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)}{2\sigma^2}\right)$$

This implies that $\mathbf{Y}$ is multivariate normal $N(\mathbf{X}\beta, \sigma^2\mathbf{I}_n)$ and the log likelihood is

$$l(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\mathbf{e}'\mathbf{e}}{2\sigma^2}$$

The maximum likelihood estimates of β and σ^2 are obtained by taking derivatives of $l(\beta, \sigma^2)$ with respect to (w.r.t) β and σ^2 and setting these partial derivatives equal to zero. Maximising $l(\beta, \sigma^2)$ w.r.t β is equivalent to minimising $\mathbf{e}'\mathbf{e}$. These beta estimates are known as the ordinary least squares (OLS) estimates.

Next we consider minimising $\mathbf{e}'\mathbf{e}$.

$$\mathbf{e}'\mathbf{e} = \sum_{j=1}^n e_j^2 = \sum_{j=1}^n (Y_j - \beta_0 - \beta_1 X_{j1} - \dots - \beta_p X_{jp})^2$$

Taking derivatives w.r.t. β_i and setting them equal to zero, we obtain

$$\begin{aligned} \frac{\delta(\mathbf{e}'\mathbf{e})}{\delta\beta_0} &= 2 \sum_{j=1}^n (Y_j - \beta_0 - \beta_1 X_{j1} - \dots - \beta_p X_{jp})(-1) = 0 \\ \frac{\delta(\mathbf{e}'\mathbf{e})}{\delta\beta_1} &= 2 \sum_{j=1}^n (Y_j - \beta_0 - \beta_1 X_{j1} - \dots - \beta_p X_{jp})(-X_{j1}) = 0 \\ \frac{\delta(\mathbf{e}'\mathbf{e})}{\delta\beta_i} &= 2 \sum_{j=1}^n (Y_j - \beta_0 - \beta_1 X_{j1} - \dots - \beta_p X_{jp})(-X_{ji}) = 0 \\ \frac{\delta(\mathbf{e}'\mathbf{e})}{\delta\beta_p} &= 2 \sum_{j=1}^n (Y_j - \beta_0 - \beta_1 X_{j1} - \dots - \beta_p X_{jp})(-X_{jp}) = 0 \end{aligned}$$

After some rearranging we have

$$\begin{aligned} n\beta_0 + \beta_1 \sum_{j=1}^n X_{j1} + \dots + \beta_p \sum_{j=1}^n X_{jp} &= \sum_{j=1}^n Y_j \\ \beta_0 \sum_{j=1}^n X_{j1} + \beta_1 \sum_{j=1}^n X_{j1}^2 + \dots + \beta_p \sum_{j=1}^n X_{j1} X_{jp} &= \sum_{j=1}^n X_{j1} Y_j \\ \beta_0 \sum_{j=1}^n X_{jp} + \beta_1 \sum_{j=1}^n X_{j1} X_{jp} + \dots + \beta_p \sum_{j=1}^n X_{jp}^2 &= \sum_{j=1}^n X_{jp} Y_j \end{aligned}$$

which can be represented in matrix notation as

$$\mathbf{X}'\mathbf{X}\beta = \mathbf{X}'\mathbf{Y} \quad (3)$$

Equations (3) are called the **Normal Equations**. If $\mathbf{X}$ is of rank k , $\mathbf{X}'\mathbf{X}$, and $(\mathbf{X}'\mathbf{X})^{-1}$ exists such that the solution (maximum likelihood estimate or MLE) to the normal equations are

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \quad (4)$$

Since $\mathbf{Y}$ is a random variable, $\hat{\beta}$ also has to be a random variable.

Theorem 7 *The maximum likelihood estimate $\hat{\beta}$ is distributed $N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. ■*

The maximum likelihood estimate of σ^2 is found by solving for σ^2 in the following equation

$$\frac{\delta l(\beta, \sigma^2)}{\delta \sigma^2} = -\frac{1}{2}n\frac{1}{\sigma^2} - \left(-\frac{1}{2}\right)\frac{1}{(\sigma^2)^2}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) = 0$$

which gives

$$\hat{\sigma}^2 = \frac{1}{n}(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta})$$

where we use the MLE $\hat{\beta}$ as an estimate for β .

Theorem 8 *The MLE $\hat{\sigma}^2$ is a biased estimate of σ^2 . ■*

To find an unbiased estimate, let

$$\begin{aligned} s^2 &= \frac{1}{n-k}(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) \\ &= \frac{1}{n-k}(\mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y}) \end{aligned}$$

In future we will always use s^2 as the estimate for σ^2 .

A continuation of the Toluca Company Example

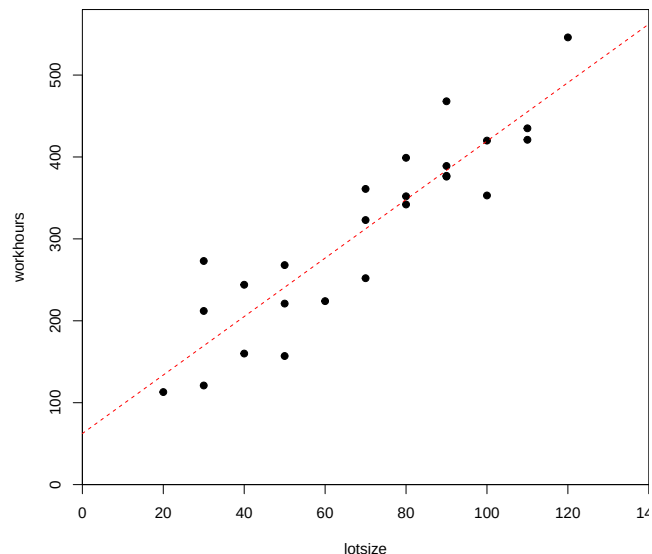
```
Y      <- as.matrix(workhours)
X      <- cbind(1, as.matrix(lotsize))
(bhat  <- solve(t(X) %*% X) %*% t(X) %*% Y)

##           [,1]
## [1,] 62.365859
## [2,]  3.570202
```

The MLE estimates for this example are $\hat{\beta} = (62.366, 3.570)'$. The first element is the intercept term while the second element is the responsiveness of work hours to lot size. i.e. as the lot size increases by 1 size, the work hours increases by 3.57 hours. The fitted line (the red dotted line) is thus, $\hat{Y} = 62.366 + 3.570X$.

Notice that some points lie close to the line while others do not. The estimated residual vector is $\hat{e} = Y - \hat{Y}$. Some residuals are positive while others are negative. The MLE estimates ensure that we have $\sum_{i=1}^n \hat{e}_i = 0$. (This result follows directly when we substitute $\hat{\beta}$ into $\frac{\delta(e'e)}{\delta\beta_0}$.)

```
par(mar = c(4,4,0,0))
plot(lotsize, workhours, pch=19,
     xlim=c(0,140), ylim=c(0,580),
     xaxs="i", yaxs="i")
abline(lm(workhours~lotsize),
       col='red', lty=2)
```



Example 2

In the following example we simulate two variables, x and y (R code shown below), each having 100 observations. $x \sim N(0,1)$ and $y = 5 + 10x + e$ where $e \sim N(0,100)$. x and e are independent of each other. The OLS β estimates are 3.972 and 9.48.

```

set.seed(1) # to be able to reproduce results from random generator

x = rnorm(100, 0, 1); # draw 100 x variables from N(0,1)
y <- 5 + 10*x + rnorm(100, 0, 10) # Use it to create a y variable
df = data.frame(x=x, y = y) # join x and y in a data frame

par(mar = c(4,4,3,0), mfrow = c(2,2)) # aesthetics of plot, margin and 2 by 2

# No use the initial df to plot
with(df, plot(x, y, main = "(a)"))

# We are randomly sampling from the 100 x and y combinations

df2 <- df[sample(100,100, replace = T),]
with(df2, points(x, y, col = 'red', pch = 19))
with(df2, abline(lm(y~x), col = 'red'))

with(df, plot(x, y, main = "(b)"))
for (i in 1:1000){abline(lm(y~x, df[sample(100,100, replace = T),]), col = 'black')}

dfint = array(dim=c(1000)); dfslope = array(dim=c(1000))
for (i in 1:1000){
dfint[i] = lm(y~x, df[sample(100,100, replace = T),])$coefficients[[1]]
dfslope[i] =lm(y~x, df[sample(100,100, replace = T),])$coefficients[[2]]
}

# We ultimately have a distribution of coefficients

hist(dfint,xlab = "int", main="(c)", prob=T)
hist(dfslope,xlab = "slope", main="(d)", prob=T)
m=mean(dfslope); s=sd(dfslope); rr=seq(6,13,length.out=150)
lines(rr,dnorm(rr,m,s),col='red')

```

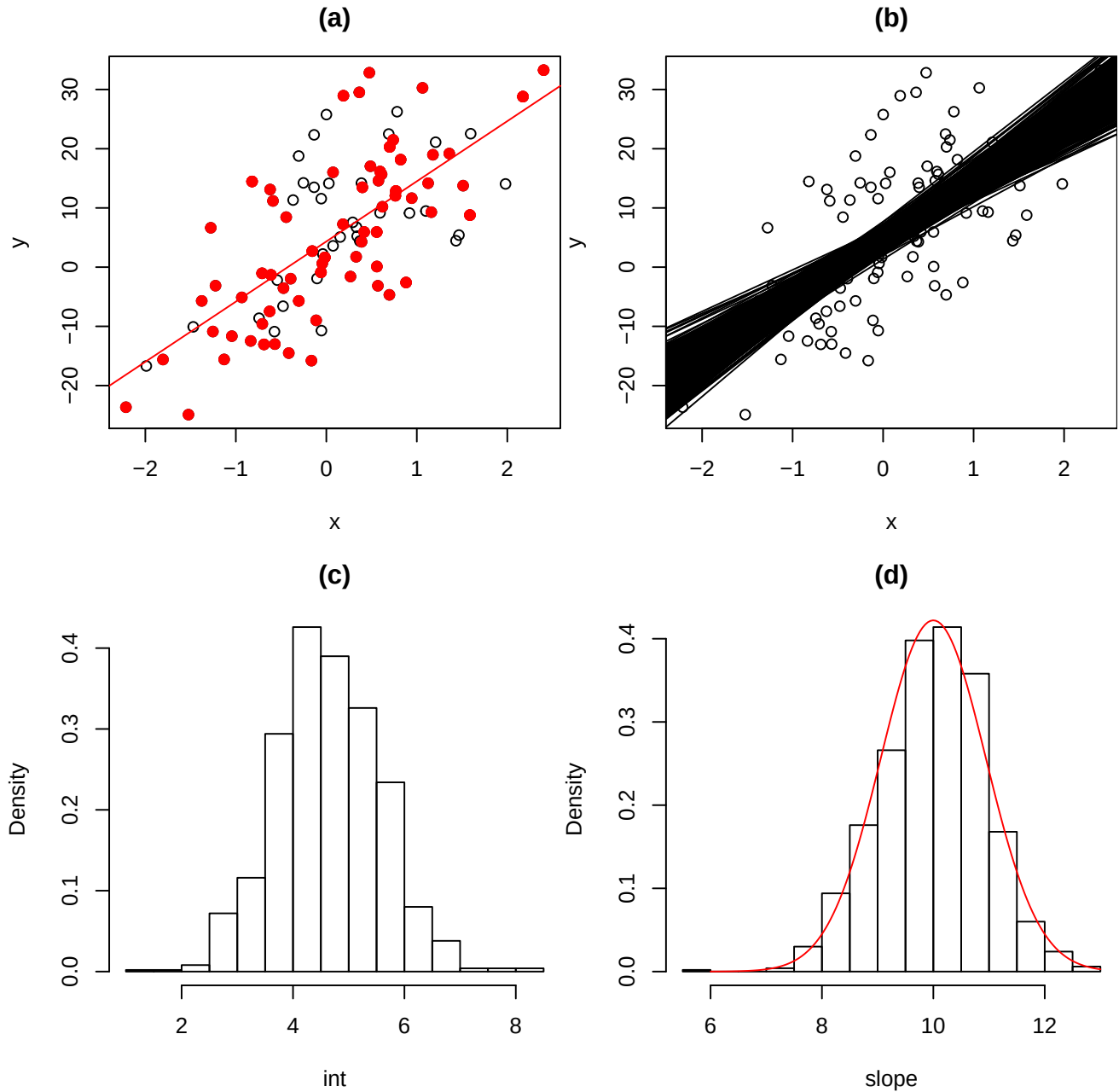


Figure (a) plots the x and y data points. We now select 100 points **with replacement** and estimate the beta parameters. The selected points as well the fitted line is indicated in (a). We now repeat the previous step 1000 times. The different fitted lines are shown in (b). From this we can see that the slope and the intercepts are different for different samples. The histograms of the different intercepts and slopes are shown in (c) and (d). Notice also that the histograms are very close to a normal distribution. The respective means and standard deviations are 4.635 and 10; and 0.935, 0.945 respectively. As we draw more and more different samples the means will tend towards the OLS estimates.

3.4 Some Distributional results based on the MLE's

Since $\hat{\beta}$ is multivariate normally distributed we can immediately write down the distributions of the marginals or linear combinations of $\hat{\beta}$. In the previous example we saw that both the intercept and the slope estimates followed a normal distribution.

Let $\beta = \begin{pmatrix} \beta^{(1)} \\ \beta^{(2)} \end{pmatrix}$ with $\beta^{(1)}$ ($q \times 1$) and $\beta^{(2)}$ ($r \times 1$), $q + r = k$ and if

$$(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{C} = \begin{pmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{pmatrix}$$

is partitioned conformably, then

$$\begin{aligned} \hat{\beta}^{(1)} &\sim N(\beta^{(1)}, \sigma^2 \mathbf{C}_{11}) \\ \hat{\beta}^{(2)} &\sim N(\beta^{(2)}, \sigma^2 \mathbf{C}_{22}) \end{aligned}$$

In particular if matrix $\mathbf{C} = (c_{ij})$ for $i, j = 1, \dots, k$ then the scalar

$$\hat{\beta}_i \sim N(\beta_i, \sigma^2 c_{ii})$$

For linear combinations of the type $\mathbf{L}\hat{\beta}$ where $\mathbf{L}$ is a $(g \times k)$ matrix of rank g then

$$\mathbf{L}\hat{\beta} \sim N(\mathbf{L}\beta, \sigma^2 \mathbf{LCL}')$$

Of particular interest is when $\mathbf{L}$ is a vector, say $\mathbf{l} = (l_0, l_1, \dots, l_p)'$, then

$$\mathbf{l}'\hat{\beta} = l_0\hat{\beta}_0 + l_1\hat{\beta}_1 + \dots + l_p\hat{\beta}_p \sim N(\mathbf{l}'\beta, \sigma^2 \mathbf{l}'\mathbf{C}\mathbf{l})$$

Notice that $\mathbf{l}'\beta$ is a scalar while $\mathbf{L}\beta$ is a matrix. If $\mathbf{l}' = (0, \dots, 1, \dots, 0)$ with the 1 in the i^{th} position, then $\mathbf{l}'\beta = \beta_i$.

A continuation of Example 2

```
x <- cbind(1, as.matrix(x))
t(x) %*% x

##           [,1]      [,2]
## [1,] 100.00000 10.88874
## [2,] 10.88874 81.05509

(C = solve(t(x) %*% x))

##           [,1]      [,2]
## [1,] 0.010148448 -0.001363317
## [2,] -0.001363317 0.012520432
```

Using the simulated data we have

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 0.01015 & -0.00136 \\ -0.00136 & 0.01252 \end{pmatrix}$$

such that

$$\begin{aligned}\hat{\beta}^{(int)} &\sim N(\beta^{(int)}, 0.01015\sigma^2) \\ \hat{\beta}^{(slope)} &\sim N(\beta^{(slope)}, 0.01252\sigma^2) \\ \hat{\beta}^{(int)} + \hat{\beta}^{(slope)} &\sim N(\beta^{(int)} + \beta^{(slope)}, 0.01994\sigma^2)\end{aligned}$$

The final equation follows from

```
L = cbind(1, 1)
L %*% C %*% t(L)

##           [,1]
## [1,] 0.01994225
```

It is important to notice that the beta estimates are unbiasedly estimated.

Under normal theory we now derive the distribution of s^2 .

Theorem 9 In the model $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ where $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$

$\frac{n-k}{\sigma^2} s^2$ is distributed as a χ^2 variate with $n-k$ degrees of freedom

Theorem 10 If $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ with $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ then

$$\hat{\beta} \text{ and } \frac{n-k}{\sigma^2} s^2$$

are independently distributed. ■

4 Confidence Intervals

Let the linear model be given by $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ with $\mathbf{e} \sim \mathbf{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. The mean of Y is

$$E(Y) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = \mathbf{x}'\beta$$

where $\mathbf{x}' = (1, X_1, \dots, X_p)$. Let $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}$ then

$$\mathbf{x}'\hat{\beta} \sim N(\mathbf{x}'\beta, \sigma^2 \mathbf{x}'\mathbf{C}\mathbf{x})$$

Thus

$$\frac{\mathbf{x}'\hat{\beta} - \mathbf{x}'\beta}{\sqrt{\sigma^2 \mathbf{x}'\mathbf{C}\mathbf{x}}} \sim N(0, 1) \text{ and independently of } \frac{n-k}{\sigma^2} s^2 \sim \chi_{n-k}^2$$

such that

$$t = \frac{\frac{\mathbf{x}'\hat{\beta} - \mathbf{x}'\beta}{\sqrt{\sigma^2 \mathbf{x}'\mathbf{C}\mathbf{x}}}}{\sqrt{\frac{n-k}{\sigma^2} s^2 \frac{1}{n-k}}} = \frac{\mathbf{x}'\hat{\beta} - \mathbf{x}'\beta}{\sqrt{s^2 \mathbf{x}'\mathbf{C}\mathbf{x}}} \sim t_{n-k} \quad (5)$$

We can now construct confidence intervals of $E(\mathbf{Y})$ by using equation (5).

Let

$$\begin{aligned} P(-t_{\alpha/2} \leq t \leq t_{\alpha/2}) &= (1 - \alpha) \\ &= P\left(-t_{\alpha/2} \leq \frac{\mathbf{x}'\hat{\beta} - \mathbf{x}'\beta}{\sqrt{s^2 \mathbf{x}'\mathbf{C}\mathbf{x}}} \leq t_{\alpha/2}\right) \\ &= P\left(\mathbf{x}'\hat{\beta} - t_{\alpha/2} \sqrt{s^2 \mathbf{x}'\mathbf{C}\mathbf{x}} \leq \mathbf{x}'\beta \leq \mathbf{x}'\hat{\beta} + t_{\alpha/2} \sqrt{s^2 \mathbf{x}'\mathbf{C}\mathbf{x}}\right) \end{aligned}$$

A $(1 - \alpha)$ confidence interval for $E(Y)$ is thus

$$\mathbf{x}'\hat{\beta} \pm t_{\alpha/2} \sqrt{s^2 \mathbf{x}'\mathbf{C}\mathbf{x}} \quad (6)$$

For any other linear combination $\mathbf{l}'\beta$ where $\mathbf{l}' = (l_0, l_1, \dots, l_p)$, $\mathbf{l}'\hat{\beta} \sim N(\mathbf{l}'\beta, \sigma^2 \mathbf{l}'\mathbf{C}\mathbf{l})$. Thus a $(1 - \alpha)$ confidence interval for the linear combination $\mathbf{l}'\beta$ is

$$\mathbf{l}'\hat{\beta} \pm t_{\alpha/2} \sqrt{s^2 \mathbf{l}'\mathbf{C}\mathbf{l}}$$

In particular

$$\beta_i \in \hat{\beta}_i \pm t_{\alpha/2} \sqrt{s^2 c_{ii}} \quad (7)$$

We would like the confidence intervals given by equation (7) to be narrow (small standard errors $\sqrt{s^2 c_{ii}}$). If some of the confidence intervals are very wide compared to others, some c_{ii} values might be relatively large, implying the possibility of severe collinearities in the data matrix $\mathbf{X}$. Collinearity often occurs if the design matrix ($\mathbf{X}$) contains many highly correlated variables. The calculation of $(\mathbf{X}'\mathbf{X})^{-1}$ may then be very inaccurate and the OLS estimates may be unreliable. Corrective remedies are strongly advised. Refer to Thiart (1990) for more details.

Example 3: The Bank Data

The example uses the **Bank Data**. The following figure can be used in order to identify significant regressor variables. Initially we will use all 8 explanatory variables. We see that Y appears to be linearly related to X_1 , X_2 , X_3 , X_6 , X_7 and X_8 . Notice also that many of the explanatory variables also appear to be correlated, For example, X_2 and X_3 or X_6 , X_7 and X_8 . When this happens it is best to exclude some of the variables from the analysis. $n = 48$ and $s = 13.02$.

```
library(car); bankdf <- read.csv("bank.csv", header = T)

## Loading required package: carData

modbank <- lm(Y~., bankdf)
beta    <- modbank$coefficients
(s      <- summary(modbank)$sigma)

## [1] 13.02213

scatterplotMatrix(bankdf)
```

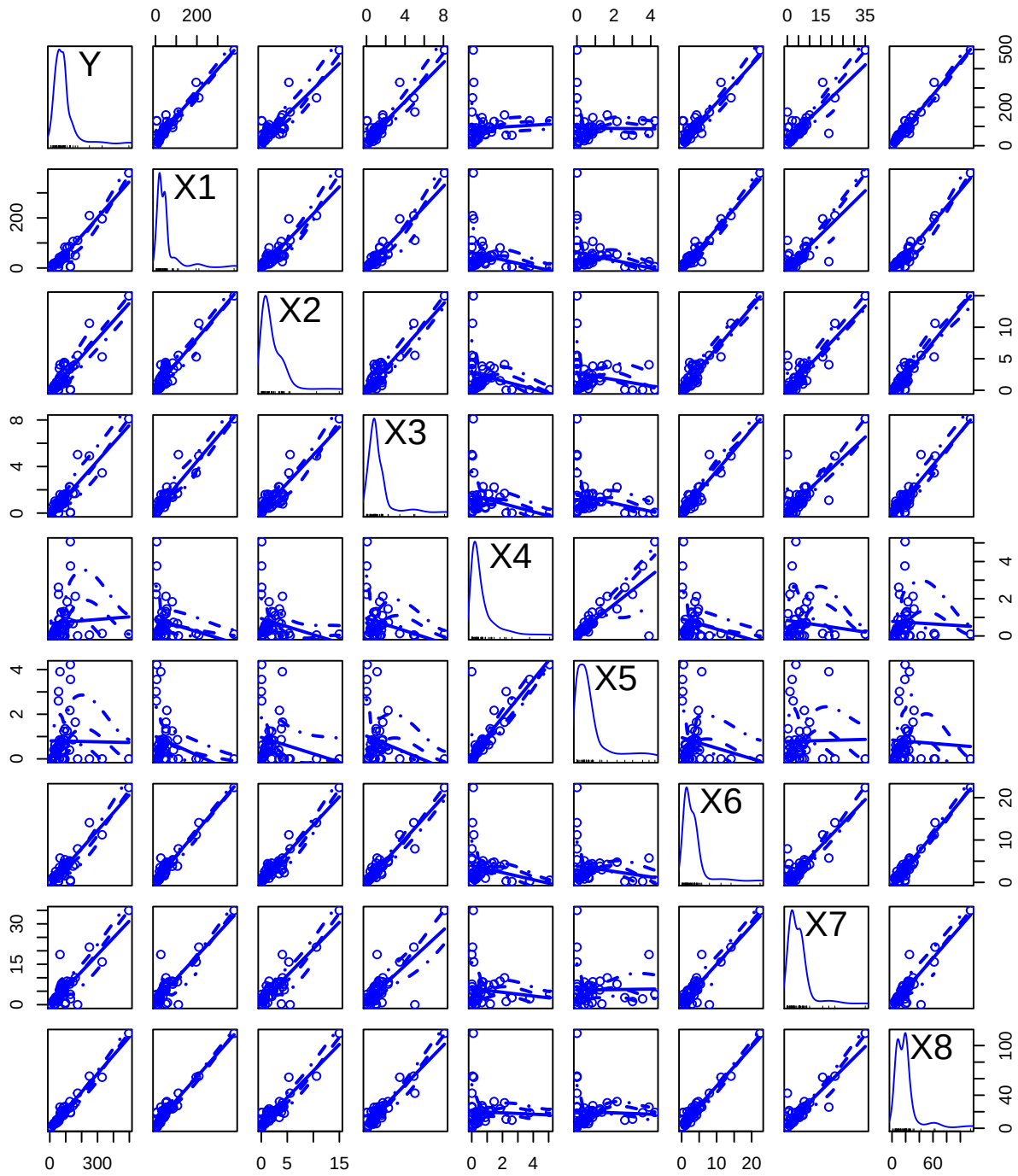


Figure 1: Scatterplot of the Bank Data

The estimation results (β estimates and the standard errors) are displayed below

```
xtable::xtable(modbank)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.2031	3.2386	0.99	0.3287
X1	0.5435	0.1721	3.16	0.0031
X2	-5.2677	2.2554	-2.34	0.0247
X3	0.8175	4.9872	0.16	0.8706
X4	11.5285	4.1293	2.79	0.0081
X5	-3.2608	4.3362	-0.75	0.4566
X6	-4.5805	2.8960	-1.58	0.1218
X7	-0.1839	0.8848	-0.21	0.8365
X8	4.1591	0.6533	6.37	0.0000

Suppose $\mathbf{x} = (1, 1, 1, 1, 1, 1, 1, 1, 1)'$ and using $(\mathbf{X}'\mathbf{X})^{-1}$ displayed below,

```
x      <- c(rep(1,9))
Xm     <- cbind(1, as.matrix(bankdf[,c(2:9)]))
(C     <- round(solve(t(Xm) %*% Xm), 4))
```

```
##           X1      X2      X3      X4      X5      X6      X7      X8
##      0.0619  0.0005 -0.0003 -0.0156 -0.0103 -0.0060  0.0015 -0.0007 -0.0017
## X1  0.0005  0.0002  0.0002 -0.0002 -0.0009  0.0027 -0.0009 -0.0002 -0.0004
## X2 -0.0003  0.0002  0.0300 -0.0127  0.0090  0.0045 -0.0009 -0.0029 -0.0028
## X3 -0.0156 -0.0002 -0.0127  0.1467  0.0120  0.0022 -0.0380  0.0128 -0.0042
## X4 -0.0103 -0.0009  0.0090  0.0120  0.1006 -0.0749  0.0222  0.0066 -0.0053
## X5 -0.0060  0.0027  0.0045  0.0022 -0.0749  0.1109 -0.0144 -0.0086 -0.0039
## X6  0.0015 -0.0009 -0.0009 -0.0380  0.0222 -0.0144  0.0495 -0.0047 -0.0027
## X7 -0.0007 -0.0002 -0.0029  0.0128  0.0066 -0.0086 -0.0047  0.0046 -0.0004
## X8 -0.0017 -0.0004 -0.0028 -0.0042 -0.0053 -0.0039 -0.0027 -0.0004  0.0025
```

then

$$\mathbf{x}'\hat{\beta}=6.9589$$

```
t(x) %*% beta
```

```
##           [, 1]
## [1, ] 6.95891
```

$$E(Y) \in 6.9589 \pm t_{39}^{(0.025)}(6.2193)$$

```
s*sqrt(t(x) %*% C %*% x)
```

```
##           [,1]
## [1,] 6.219345
```

$$\beta_1 \in 0.5435 \pm t_{39}^{(0.025)}(0.1721)$$

```
s*sqrt(solve(t(Xm) %*% Xm)[2,2]) # No rounding on C
```

```
## [1] 0.1721339
```

```
s*sqrt(C[2,2]) # With rounding on C
```

```
## [1] 0.1841607
```

$$\beta_2 + \beta_5 \in -8.5285 \pm t_{39}^{(0.025)}(5.0418)$$

```
x <- c(0,0,1,0,0,1,0,0,0)
t(x) %*% beta
```

```
##           [,1]
## [1,] -8.528503
```

```
s*sqrt(t(x) %*% C %*% x)
```

```
##           [,1]
## [1,] 5.041768
```

For any general linear combination $\mathbf{L}\beta$ where $\mathbf{L}$ ($q \times k$) is of rank q , $\mathbf{L}\hat{\beta}$ has the distribution- $N(\mathbf{L}\beta, \sigma^2 \mathbf{LCL}')$. To set up a *confidence region* for $\mathbf{L}\beta$ we need the following result.

Theorem 11 If $\mathbf{Y}$ ($q \times 1$) has a multivariate normal distribution $N(\mathbf{0}, \Sigma)$ then $\mathbf{Y}'\Sigma^{-1}\mathbf{Y}$ is distributed χ_q^2 . ■

Since $\hat{\beta} - \beta \sim N(\mathbf{0}, \sigma^2 \mathbf{C})$ it follows from the previous theorem that

$$(\hat{\beta} - \beta)'(\sigma^2 \mathbf{C})^{-1}(\hat{\beta} - \beta) \sim \chi_k^2$$

and is also independent of $\frac{n-k}{\sigma^2}s^2 \sim \chi_{n-k}^2$. Thus the ratio

$$F = \frac{\frac{(\hat{\beta}-\beta)'(\sigma^2\mathbf{C})^{-1}(\hat{\beta}-\beta)}{k}}{\frac{\frac{n-k}{\sigma^2}s^2 \frac{1}{n-k}}{ks^2}} = \frac{(\hat{\beta}-\beta)'(\mathbf{C})^{-1}(\hat{\beta}-\beta)}{ks^2} \sim F_{k,n-k} \quad (8)$$

If the $P(F \leq F_\alpha) = (1 - \alpha)$ then

$$P\left(\frac{(\hat{\beta}-\beta)'(\mathbf{C})^{-1}(\hat{\beta}-\beta)}{ks^2} \leq F_\alpha\right) = P((\hat{\beta}-\beta)'(\mathbf{C})^{-1}(\hat{\beta}-\beta) \leq ks^2 F_\alpha) = (1 - \alpha)$$

and a $(1 - \alpha)$ confidence region for β is given by

$$(\hat{\beta}-\beta)'(\mathbf{C})^{-1}(\hat{\beta}-\beta) \leq ks^2 F_\alpha(k, n-k) \quad (9)$$

For any linear combination $\mathbf{L}\beta$ we have $\mathbf{L}\hat{\beta} - \mathbf{L}\beta \sim N(\mathbf{0}, \sigma^2(\mathbf{LCL}'))$ so that

$$(\mathbf{L}\hat{\beta} - \mathbf{L}\beta)'(\sigma^2\mathbf{LCL}')^{-1}(\mathbf{L}\hat{\beta} - \mathbf{L}\beta) \sim \chi_q^2 \quad (10)$$

provided the matrix $\mathbf{L}$ ($q \times k$) is of rank q . Since equation (10) is also independent of $\frac{n-k}{\sigma^2}s^2 \sim \chi_{n-k}^2$, a $(1 - \alpha)$ confidence region for $\mathbf{L}\beta$ is given by

$$(\mathbf{L}\hat{\beta} - \mathbf{L}\beta)'(\mathbf{LCL}')^{-1}(\mathbf{L}\hat{\beta} - \mathbf{L}\beta) \leq qs^2 F_\alpha(q, n-k)$$

Note that if

$$\beta = \begin{pmatrix} \beta^{(1)} \\ \beta^{(2)} \end{pmatrix}, \quad \beta^{(1)} (q \times 1), \quad \beta^{(2)} (r \times 1), \quad q + r = k$$

then

$$\begin{aligned} (\hat{\beta}^{(1)} - \beta^{(1)})'(\sigma^2 C_{11})^{-1}(\hat{\beta}^{(1)} - \beta^{(1)}) &\sim \chi_q^2 \\ (\hat{\beta}^{(2)} - \beta^{(2)})'(\sigma^2 C_{22})^{-1}(\hat{\beta}^{(2)} - \beta^{(2)}) &\sim \chi_r^2 \end{aligned}$$

Example 4: confidence intervals of forecasts

We can also set a confidence interval on a future value of the response Y . As an example let Y = Average first year mark and X = a students Matric mathematics mark. Assume that (Y, X) follows a bivariate normal distribution with $E(Y|x) = \beta_0 + \beta_1 x$.

Let the value (future) for the X variable be $X'_f = (1, x_{1f}, x_{2f}, \dots, x_{pf})$. We assume that X_f is known. Let the actual future value of the response be Y_f which we **do not know**. Let the predicted value be $\hat{Y}_f$ where $\hat{Y}_f$ is computed from our estimated $\hat{\beta}$ and s^2 as follows -

$$\hat{Y}_f = \hat{\beta}_0 + \hat{\beta}_1 x_{1f} + \dots + \hat{\beta}_p x_{pf} = X'_f \hat{\beta}$$

Since the actual, but unknown value of the response is Y_f , consider the difference $Z = Y_f - \hat{Y}_f$ with

$$E(Z) = E(Y_f - \hat{Y}_f) = E(Y_f) - E(\hat{Y}_f) = X'_f \beta - X'_f E(\hat{\beta}) = \mathbf{X}'_f \beta - \mathbf{X}'_f \beta = 0$$

The variance of Z is

$$\text{var}(Z) = \text{var}(Y_f - \hat{Y}_f) = \text{var}(Y_f) + \text{var}(\hat{Y}_f) = \sigma^2 + \sigma^2(X'_f \mathbf{C} X_f) \quad (11)$$

Thus

$$\frac{Z}{\sqrt{\sigma^2(1 + X'_f \mathbf{C} X_f)}} = \frac{Y_f - \hat{Y}_f}{\sqrt{\sigma^2(1 + X'_f \mathbf{C} X_f)}} \sim N(0, 1) \text{ independent of } \frac{n-k}{\sigma^2} s^2 \sim \chi^2_{n-k}$$

and

$$t = \frac{Y_f - \hat{Y}_f}{\sqrt{s^2(1 + X'_f \mathbf{C} X_f)}} \sim t_{n-k}$$

such that a $(1 - \alpha)$ prediction interval for Y_f is

$$Y_f \in \hat{Y}_f \pm t_{\alpha/2} \sqrt{s^2(1 + X'_f \mathbf{C} X_f)} = X'_f \hat{\beta} \pm t_{\alpha/2} \sqrt{s^2(1 + X'_f \mathbf{C} X_f)} \quad (12)$$

This confidence interval is slightly wider than the confidence interval found in equation (6).

Suppose we have several future observations, say m of them. Let $\bar{Y}_f$ -be the mean of them. Then a prediction value for a future $\bar{Y}_f$ will be

$$\bar{Y}_f \in \hat{Y}_f \pm t_{\alpha/2} \sqrt{s^2 \left(\frac{1}{m} + X'_f \mathbf{C} X_f \right)} = X'_f \hat{\beta} \pm t_{\alpha/2} \sqrt{s^2 \left(\frac{1}{m} + X'_f \mathbf{C} X_f \right)} \quad (13)$$

5 Tests of Hypotheses

Using the previous distributional results we can test any hypothesis on the β 's including subsets of the β 's, individual β 's, and any linear combination of the β 's. We can also test hypotheses about σ^2 . In the following chapter we have chosen to start by testing for the significance of single β_i 's and then end off with the introduction of the Analysis of variance table (ANOVA). Most other text do the opposite of what was done here.

Example 5: Testing the significance of one β

Lets return to the Bank data example. We saw that $\hat{\beta}_3 = 0.8175$ with a standard deviation of 4.9872. We can formally test whether $\beta_3 = 0$ (while the other variables are assumed to be in the model) as follows:

$$\begin{aligned} H_0 &: \beta_3 = 0 \text{ against} \\ H_1 &: \beta_3 \neq 0 \end{aligned}$$

We know that $\hat{\beta}_3 \sim N(\beta_3, \sigma^2 c_{33})$ independently of $\frac{39}{\sigma^2} s^2 \sim \chi_{39}^2$. Note that here we are assuming that $\mathbf{C}$ has diagonal elements $c_{00}, \dots, c_{88}$. Assuming that H_0 is true then

$$Z = \frac{\hat{\beta}_3}{\sqrt{\sigma^2 c_{33}}} \sim N(0, 1)$$

but since we do not know σ^2 in general we cannot use Z as a test statistic. We can however use a t statistic as follows

$$t = \frac{\frac{\hat{\beta}_3}{\sqrt{\sigma^2 c_{33}}}}{\sqrt{\frac{s^2}{\sigma^2}}} = \frac{\hat{\beta}_3}{\sqrt{s^2 c_{33}}} \sim t_{39}$$

In this case $t = 0.163$ (i.e. $\frac{\hat{\beta}_3}{\text{std}(\hat{\beta}_3)}$) with a p value of 0.87 which suggests that we cannot reject H_0 . Note that we could have used a F test as well since

$$\hat{\beta}_3' (\sigma^2 c_{33})^{-1} \hat{\beta}_3 = \frac{\hat{\beta}_3^2}{\sigma^2 c_{33}} \sim \chi_1^2$$

such that

$$\frac{\frac{\hat{\beta}_3^2}{\sigma^2 c_{33}}}{\frac{s^2}{\sigma^2}} = \frac{\hat{\beta}_3^2}{s^2 c_{33}} = \left(\frac{\hat{\beta}_3}{\sqrt{s^2 c_{33}}} \right)^2 = (t_{39})^2 \sim F_{1,39}$$

Example 6: Testing the significance of a subset of β 's

After reexamining the betas and the standard errors of the Bank data example it appears as if the $\hat{\beta}_0, \hat{\beta}_3, \hat{\beta}_5, \hat{\beta}_6$ and $\hat{\beta}_7$ are all 0. Let $\beta = \begin{pmatrix} \beta^{(1)} & \beta^{(2)} \end{pmatrix}'$ where

$$\begin{aligned}\beta^{(1)} &= (\beta_0, \beta_3, \beta_5, \beta_6, \beta_7)' \\ \beta^{(2)} &= (\beta_1, \beta_2, \beta_4, \beta_8)'\end{aligned}$$

We now partition $\mathbf{C}$ such that $\mathbf{C}_{11}$ contains all of the elements of $\mathbf{C}$ associated with $\beta_0, \beta_3, \beta_5, \beta_6$ and β_7 . Similarly $\mathbf{C}_{22}$ contains all of the elements of $\mathbf{C}$ associated with $\beta_1, \beta_2, \beta_4$ and β_8 . $\mathbf{C}_{11}$ is displayed below

```
(C11 <- solve(t(Xm) %*% Xm)[c(1,4,6,7,8), c(1,4,6,7,8)])
```

```
##              X3              X5              X6              X7
##      0.0618521309 -0.015629162 -0.005958743  0.001505681 -0.0006801037
## X3 -0.0156291616  0.146674955  0.002246478 -0.038008211  0.0128149047
## X5 -0.0059587430  0.002246478  0.110878304 -0.014439449 -0.0085592363
## X6  0.0015056810 -0.038008211 -0.014439449  0.049455930 -0.0046720292
## X7 -0.0006801037  0.012814905 -0.008559236 -0.004672029  0.0046163286
```

We can now test

$$\begin{aligned}H_0 &: \beta^{(1)} = \mathbf{0} \text{ against} \\ H_1 &: \text{Any of the } \beta_i \in \beta^{(1)} \text{ is not equal to zero.}\end{aligned} \tag{14}$$

Assuming H_0 is true, $\hat{\beta}^{(1)} \sim N(\mathbf{0}, \sigma^2 \mathbf{C}_{11})$ independently of $\frac{39}{\sigma^2} s^2 \sim \chi_{39}^2$ such that

$$F = \frac{\hat{\beta}^{(1)'} \mathbf{C}_{11}^{-1} \hat{\beta}^{(1)}}{5s^2} \sim F_{5,39} \tag{15}$$

We will reject H_0 in favour of H_1 if $F \geq F_{5,39}^{(\alpha)}$ where $P(F \geq F_{5,39}^{(\alpha)}) = \alpha$. In this example $F = 1.2776$ and $F_{5,39}^{(0.05)} = 2.4458$ suggesting that we cannot reject H_0 .

```
beta1<- beta[c(1,4,6,7,8)] ; F = (t(beta1) %*% solve(C11) %*% beta1)/(5*s^2)
```

The above example can be generalised to test for any subset of restrictions. In this case

$$F = \frac{\hat{\beta}^{(1)'} \mathbf{C}_{11}^{-1} \hat{\beta}^{(1)}}{qs^2} \sim F_{q,n-k}$$

where $q = \dim(\hat{\beta}^{(1)})$ = the number of restrictions and k is equal to the number of explanatory variables included in the fitted model plus one (assuming that the intercept is included in the fitted model).

$\hat{\beta}^{(1)'} \mathbf{C}_{11}^{-1} \hat{\beta}^{(1)}$ could also be calculated as follows:

- Fit the restricted model (i.e. the model assuming that H_0 is true) and calculate the sums of squares due to error (SSE_R)
- Fit the unrestricted model (the full model) and calculate the sums of squares due to error (SSE_{UR})
- $\hat{\beta}^{(1)'} \mathbf{C}_{11}^{-1} \hat{\beta}^{(1)} = SSE_R - SSE_{UR}$. (A mathematical proof is not shown but one does exist.)

A continuation of Example 6

The estimation results of the restricted and the unrestricted model from the previous example is displayed below.

```
Rm = lm(formula = Y ~ X1 + X2 + X4 + X8 + 0, data = bankdf) # Restricted
uRm = lm(formula = Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8,
          data = bankdf) # Unrestricted
summary(Rm)

##
## Call:
## lm(formula = Y ~ X1 + X2 + X4 + X8 + 0, data = bankdf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -27.945  -5.833   1.398   6.673  41.430
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## X1      0.5321     0.1299   4.098 0.000177 ***
## X2     -5.9354     2.1865  -2.715 0.009443 **
## X4     11.9803     2.6335   4.549 4.20e-05 ***
## X8      3.4519     0.5133   6.725 2.89e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.23 on 44 degrees of freedom
## Multiple R-squared:  0.9894, Adjusted R-squared:  0.9885
## F-statistic: 1028 on 4 and 44 DF, p-value: < 2.2e-16
```

```
summary(uRm)

##
## Call:
## lm(formula = Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8, data = bankdf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -26.865  -7.098  -0.730   6.042  40.426
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.2031     3.2386   0.989  0.32875
## X1              0.5435     0.1721   3.158  0.00307 **
## X2             -5.2677     2.2554  -2.336  0.02475 *
## X3              0.8175     4.9872   0.164  0.87065
## X4             11.5285     4.1293   2.792  0.00807 **
## X5             -3.2608     4.3362  -0.752  0.45657
## X6             -4.5805     2.8960  -1.582  0.12180
## X7             -0.1839     0.8848  -0.208  0.83646
## X8              4.1591     0.6533   6.366 1.61e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.02 on 39 degrees of freedom
## Multiple R-squared:  0.98, Adjusted R-squared:  0.9759
## F-statistic: 238.6 on 8 and 39 DF, p-value: < 2.2e-16
```

```
(sr = summary(Rm)$sigma)

## [1] 13.22597

(nqr = summary(Rm)$df[2])

## [1] 44

(sur = summary(uRm)$sigma)

## [1] 13.02213

(nqur = summary(uRm)$df[2])

## [1] 39
```

```
(r = nqr - nqur)

## [1] 5

(Fs = (nqr*sr^2 - nqur*sur^2)/(r*sur^2))

## [1] 1.277649
```

The F statistic as calculated in Example 6 now is

$$\begin{aligned}
 F &= \frac{SSE_R - SSE_{UR}}{5(13.02^2)} \\
 &= \frac{s_R^2(n - q_R) - s_{UR}^2(n - q_{UR})}{rs_{UR}^2} \\
 &= \frac{(13.22597^2)44 - (13.02213^2)39}{5(13.02213^2)} = 1.2776^*
 \end{aligned}$$

q_R	is the number of β parameters in the restricted model
q_{UR}	is the number of β parameters in the unrestricted model
r	is the number of restrictions under H_0
n	is the sample size
s_R^2	is the residual variance of the restricted model
s_{UR}^2	is the residual variance of the unrestricted model

* the residual standard errors displayed has not been rounded, note that rounding could cause some inaccuracies in the calculation of the above F statistic and so only round the final answer.

Example 7: Should we perform regression?

With any standard regression output an F statistic is always calculated. e.g. The F statistic for the full model of the Bank data is 238.6. In this case the two models considered are

$$Y = \beta_0 + e_{\text{Restricted Model}} \quad (16)$$

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_8 X_8 + e_{\text{Unrestricted Model}} \quad (17)$$

```
summary(lm(Y~1, data = bankdf))

##
## Call:
## lm(formula = Y ~ 1, data = bankdf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -83.94 -46.44 -16.44   6.31 406.06
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    90.94      12.10   7.516 1.36e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 83.83 on 47 degrees of freedom
```

such that the F statistic used to test the hypothesis

$H_0 : \beta_1 = \beta_2 = \dots = \beta_8 = 0$ (no regression analysis is required) against

$H_1 : \text{Any of the } \beta_i \text{ } i = 1, \dots, 8 \text{ is not equal to zero.}$

is calculated as

$$F = \frac{SSE_R - SSE_{UR}}{8(13.02^2)} = \frac{(83.83^2)47 - (13.02^2)39}{8(13.02^2)} = 238.5653^*$$

5.1 The Analysis of Variance Table (ANOVA)

The F statistic used to test whether or not we should perform a regression analysis or not (as in Example 7) can also be calculated by using an ANOVA table. We saw that this F statistic has the following form

$$F = \frac{SSE_R - SSE_{UR}}{ps_{UR}^2}$$

Note that $r = p$ = the number of explanatory variables included in the full model

Let the estimated β 's from the full model be $\hat{\beta}$ and the estimated β from the restricted model be $\tilde{\beta}$ such that

$$SSE_{UR} = (\mathbf{Y} - \mathbf{X}\tilde{\beta})'(\mathbf{Y} - \mathbf{X}\tilde{\beta}) = \mathbf{Y}'\mathbf{Y} - \tilde{\beta}'\mathbf{X}'\mathbf{Y}$$

and

$$SSE_R = \left(\mathbf{Y} - \begin{pmatrix} 1 & \dots & 1 \end{pmatrix}' \tilde{\beta} \right)' \left(\mathbf{Y} - \begin{pmatrix} 1 & \dots & 1 \end{pmatrix}' \tilde{\beta} \right) = \mathbf{Y}'\mathbf{Y} - n\bar{Y}^2$$

The last result follows since the design matrix $(\tilde{X})$ only includes a column of 1's such that

$$\tilde{\beta} = (\tilde{X}'\tilde{X})^{-1} \tilde{X}'\mathbf{Y} = \bar{Y}$$

and from the normal equations for the restricted model we have

$$\begin{aligned} \tilde{X}'\tilde{X}\tilde{\beta} &= \tilde{X}'\mathbf{Y} \\ n\tilde{\beta} &= \sum_{i=1}^n Y_i \end{aligned}$$

Now

$$F = \frac{\hat{\beta}'\mathbf{X}'\mathbf{Y} - n\bar{Y}^2}{p \left(\frac{\mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y}}{n-k} \right)} = \frac{\frac{SSR}{p}}{\frac{SSE}{n-k}} = \frac{MSR}{MSE}$$

where SSR =the sums of squares due to regression, SSE =the residual sums of squares, MSR =regression mean square error and MSE =the residual mean square error. We can decompose the total sums of squares (SST) into SSR and SSE since

$$\begin{aligned} Y_i - \bar{Y} &= Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y} \\ 0 &= \sum_i (Y_i - \hat{Y}_i)(\hat{Y}_i - \bar{Y}) \end{aligned}$$

such that

$$\begin{aligned} SST &= SSR + SSE \\ \sum (Y_i - \bar{Y})^2 &= \sum (\hat{Y}_i - \bar{Y})^2 - \sum (Y_i - \hat{Y}_i)^2 \end{aligned}$$

The construction of the above F statistic can be summarised in the following table:

Source of Variation	Degrees of Freedom	Sums of squares	Mean Squares	F
Regression	$k - 1 = p$	$SSR = \hat{\beta}'\mathbf{X}'\mathbf{Y} - n\bar{Y}^2$	$\frac{SSR}{p}$	$\frac{MSR}{MSE}$
Error	$n - k$	$SSE = \mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y}$	$\frac{SSE}{n-k}$	
Total	$n - 1$	$SST = \mathbf{Y}'\mathbf{Y} - n\bar{Y}^2$	$s_y^2 = \sum \frac{(Y_i - \bar{Y})^2}{n-1}$	

Example 8: A continuation of the Bank Data Example

The output above displays the estimation results of the full model of the Bank data. It contains the values of the estimated coefficients, their standard errors ($se(\hat{\beta})$ the square root of the diagonal elements of $s^2(X'X)^{-1}$) and their t statistics $\left(t_i = \frac{\hat{\beta}_i}{se(\hat{\beta}_i)}\right)$. Significant regressors are indicated with stars (*, ** or ***). Notice also that the F statistic for significant regression is displayed. (F statistic = 238.6 on 8 and 39 degrees of freedom). The F statistic indicates that we would reject the hypothesis of no regression. $s = 13.02$, the residual standard error, with 39 degrees of freedom. From the estimation results we can see that β_1 is significantly different from 0 while β_3 is not significantly different from 0. These results are based on t tests used to test $H_0 : \beta_i = 0$ against the alternative that $\beta_i \neq 0$.

5.2 The Wald Test

The Wald test is often used in financial applications. It is a test on the restrictions of the parameters β_i of a regression model. Consider the following hypothesis tests

$$H_0 : \mathbf{L}\beta = \mathbf{l}_0 \text{ where the } q \times k \text{ matrix } \mathbf{L} \text{ is assumed to be of rank } q$$

$$H_1 : \mathbf{L}\beta \neq \mathbf{l}_0$$

Now for any linear combination $\mathbf{L}\beta$ we have $\mathbf{L}\hat{\beta} - \mathbf{L}\beta \sim N(\mathbf{0}, \sigma^2(\mathbf{LCL}'))$ so that

$$(\mathbf{L}\hat{\beta} - \mathbf{L}\beta)'(\sigma^2\mathbf{LCL}')^{-1}(\mathbf{L}\hat{\beta} - \mathbf{L}\beta) \sim \chi_q^2 \quad (18)$$

provided the matrix $\mathbf{L}$ is of rank q . Since (18) is also independent of $\frac{n-k}{\sigma^2}s^2 \sim \chi_{n-k}^2$ the required F statistic is

$$F = \frac{(\mathbf{L}\hat{\beta} - \mathbf{l}_0)'(\mathbf{LCL}')^{-1}(\mathbf{L}\hat{\beta} - \mathbf{l}_0)}{qs^2} \sim F_{q, n-k}$$

Example 9: Wald Tests

To test

$$H_0 : \beta_i = \beta_j \text{ or } \beta_i - \beta_j = 0$$

$$H_1 : \beta_i \neq \beta_j$$

use

$$\mathbf{L} = (0, 0, \dots, 1, \dots, -1, 0, \dots, 0)$$

with 1 in the i^{th} position and -1 in the j^{th} position with $l_0 = 0$

To test

$$H_0 : \beta_3 = 3$$

$$H_1 : \beta_3 \neq 0$$

use

$$\mathbf{L} = (0, 0, 0, 1, 0, \dots, 0)$$

To test

$$H_0 : \beta_1 = \beta_3, \beta_2 = \beta_4$$

$$H_1 : \beta_1 \neq \beta_3, \beta_2 \neq \beta_4$$

use

$$\mathbf{L} = \begin{pmatrix} 0 & 1 & 0 & -1 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & -1 & \dots & 0 \end{pmatrix}$$

6 The Coefficient of Determination

Consider the linear model $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ where $E(\mathbf{e}) = \mathbf{0}$, $E(\mathbf{e}\mathbf{e}') = \sigma^2\mathbf{I}_n$, that is, we make no assumption about the distribution of $\mathbf{e}$. We can show that the average value of $\hat{Y} = \hat{Y}_{ave} = \bar{Y}$. The correlation between Y and $\hat{Y}$ is defined as

$$\begin{aligned} R &= \frac{\sum_{j=1}^n (Y_j - \bar{Y})(\hat{Y}_j - \hat{Y}_{ave})}{\sqrt{\sum_{j=1}^n (Y_j - \bar{Y})^2 \sum_{j=1}^n (\hat{Y}_j - \hat{Y}_{ave})^2}} \\ &= \frac{\sum_{j=1}^n (Y_j - \bar{Y})(\hat{Y}_j - \bar{Y})}{\sqrt{\sum_{j=1}^n (Y_j - \bar{Y})^2 \sum_{j=1}^n (\hat{Y}_j - \bar{Y})^2}} \end{aligned}$$

We can show that R simplifies to

$$R = \sqrt{\frac{\sum_{j=1}^n \hat{Y}_j^2 - n\bar{Y}^2}{\sum_{j=1}^n Y_j^2 - n\bar{Y}^2}} = \sqrt{\frac{\hat{\beta}'\mathbf{X}'\mathbf{Y} - n\bar{Y}^2}{\mathbf{Y}'\mathbf{Y} - n\bar{Y}^2}}$$

Instead of R we rather use

$$R^2 = \frac{\hat{\beta}'\mathbf{X}'\mathbf{Y} - n\bar{Y}^2}{\mathbf{Y}'\mathbf{Y} - n\bar{Y}^2} = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 is called the coefficient of determinant (also called the Multiple R squared in R), $0 \leq R^2 \leq 1$. It measures the strength of the **linear** relationship between the response variable and the fitted values ($\hat{Y}$). We can also interpret R^2 as the percentage of the total variation in Y that is explained due to performing regression analysis, i.e. $\frac{SSR}{SST}$. A good or satisfactory model is when a large percentage of the variation is explained by the model. A value close to 0 indicates that a linear relationship does not exist between Y and $\hat{Y}$ while R^2 close to 1 indicates a good linear relationship. Recall that the R^2 for the bank data is 0.98 indicating a good fit.

Now $SSE \sim \sigma^2 \chi_{n-k}^2$ while $SST \sim \sigma^2 \chi_{n-1}^2$. We define the adjusted R^2 as

$$R_{adj}^2 = 1 - \frac{SSE/(n-k)}{SST/(n-1)} = 1 - (1 - R^2) \left(\frac{n-1}{n-k} \right)$$

R_{adj}^2 adjusts for the degrees of freedom of SSE and SST . This is very useful when comparing different models consisting of different β 's (that is, different $\mathbf{X}$'s).

7 Model Checking and the Analysis of Residuals

7.1 Introduction

Once we have fitted our regression model we need to check that the assumptions of the linear model are not violated. We will also have to check for the presence of outliers and influential observations and whether certain observations should be deleted or downweighted. We will thus have to check whether the following assumptions are not violated:

1. $E(e_j) = 0 \forall j$
2. $E(e_j^2) = \sigma^2 \forall j$
3. $E(e_j e_k) = 0$ for $j \neq k \forall j$
4. $e_j \sim N(0, \sigma^2) \forall j$

Assumption 1 can easily be checked by plotting the histogram of the estimated residuals or by applying t tests.

Assumption 2 implies that we have *homoscedastic errors* – the same and equal variances. If $E(e_j) = \sigma_j^2$ then we have *heteroscedastic errors*. This topic will not be covered in this course but is covered in more advanced courses.

Assumption 3 implies that the errors are un-correlated. If they are not, we say that the residuals are serially correlated errors and that they can be modelled by using time series analysis. This topic is also covered in advanced courses.

Assumption 4 implies that $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. Various tests will be discussed in order to test for departures from normality.

7.2 Estimated Residuals

The residuals (errors) e_j , $j = 1, \dots, n$, are **unknown** and they are estimated using the estimated residuals $\hat{e}_j$, $j = 1, \dots, n$. Now

$$\begin{aligned}\hat{\mathbf{e}} &= \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{Y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{Y} \\ &= (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')(\mathbf{X}\boldsymbol{\beta} + \mathbf{e}) \\ &= \mathbf{0} + (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{e} \\ &= (\mathbf{I} - \mathbf{H})\mathbf{e}\end{aligned}$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \quad (19)$$

and is called the "**Hat**" or projection matrix since

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{H}\mathbf{Y}$$

Thus the effect of $\mathbf{H}$ on $\mathbf{Y}$ gives $\hat{\mathbf{Y}}$. Now

$$\hat{\mathbf{e}} = (\mathbf{I} - \mathbf{H})\mathbf{e} = \mathbf{M}\mathbf{e} \text{ with } \mathbf{M} = \mathbf{I} - \mathbf{H} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

and $\mathbf{M}^2 = \mathbf{M}$, $\mathbf{M}' = \mathbf{M}$, i.e., $\mathbf{M}$ is an idempotent matrix. Since $E(\mathbf{e}\mathbf{e}') = \sigma^2\mathbf{I}_n$, it follows that

$$E(\hat{\mathbf{e}}\hat{\mathbf{e}}') = \mathbf{M}(\sigma^2\mathbf{I}_n)\mathbf{M}' = \sigma^2\mathbf{M}^2 = \sigma^2\mathbf{M}$$

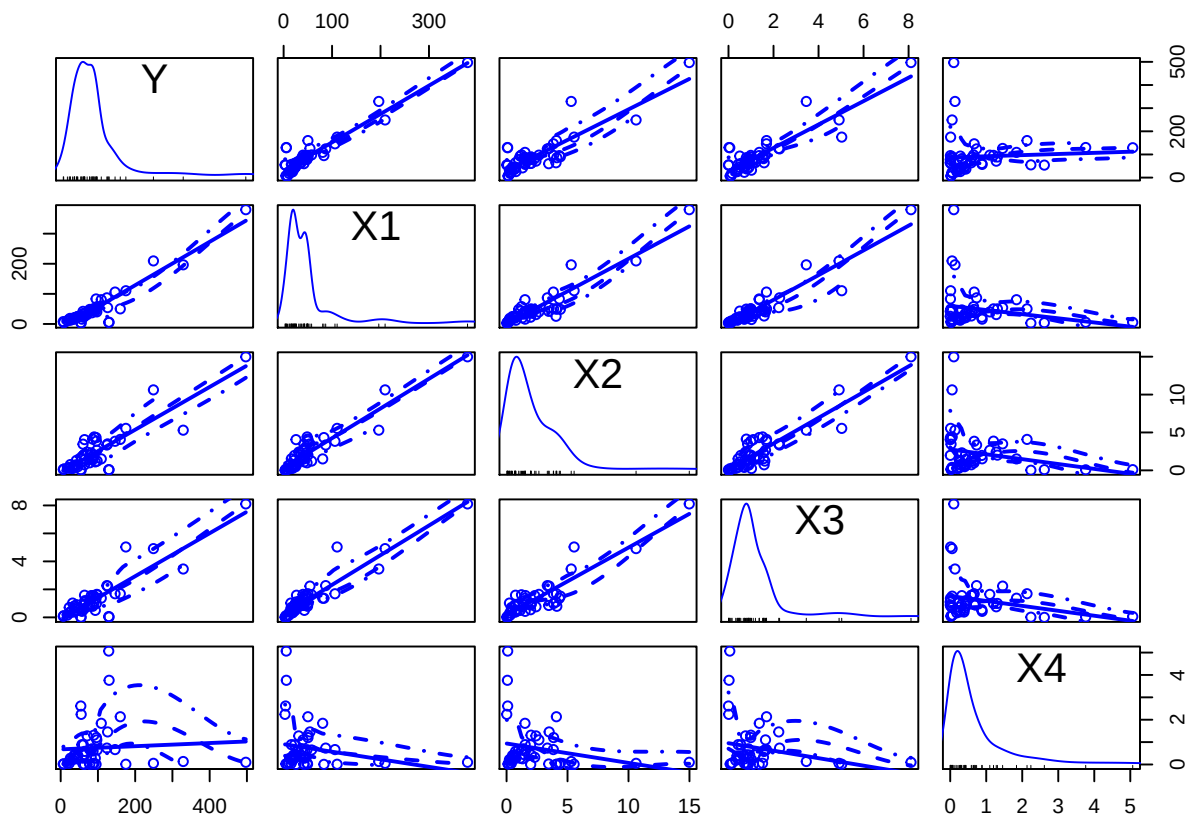
such that $(\hat{e}_1, \dots, \hat{e}_n)$ are **correlated**.

7.3 Model Checking Plots

It often useful to use plots in order to check whether the final regression model is adequate or not. Various plots can be used.

7.3.1 A Matrix Plot

```
scatterplotMatrix(bankdf[, c(1:5)] )
```



A **Matrix Plot** of the dependent variable Y and all the explanatory variables $(X_1, \dots, X_p)$ is very useful. This will show, apart from relationships between Y and the X 's, the relationships among the X - variables. These are called multi-collinearities and they can be harmful to the least squares estimation procedure. It will also further highlight outliers and/or influential observations. The above plot displays the relationships between Y , X_1 , X_2 , X_3 , and X_4 of the Bank data. Y appears to be linearly related to X_1 , X_2 and X_3 . Notice also that X_1 , X_2 and X_3 are highly correlated. The plots also displays the presence of possible outliers.

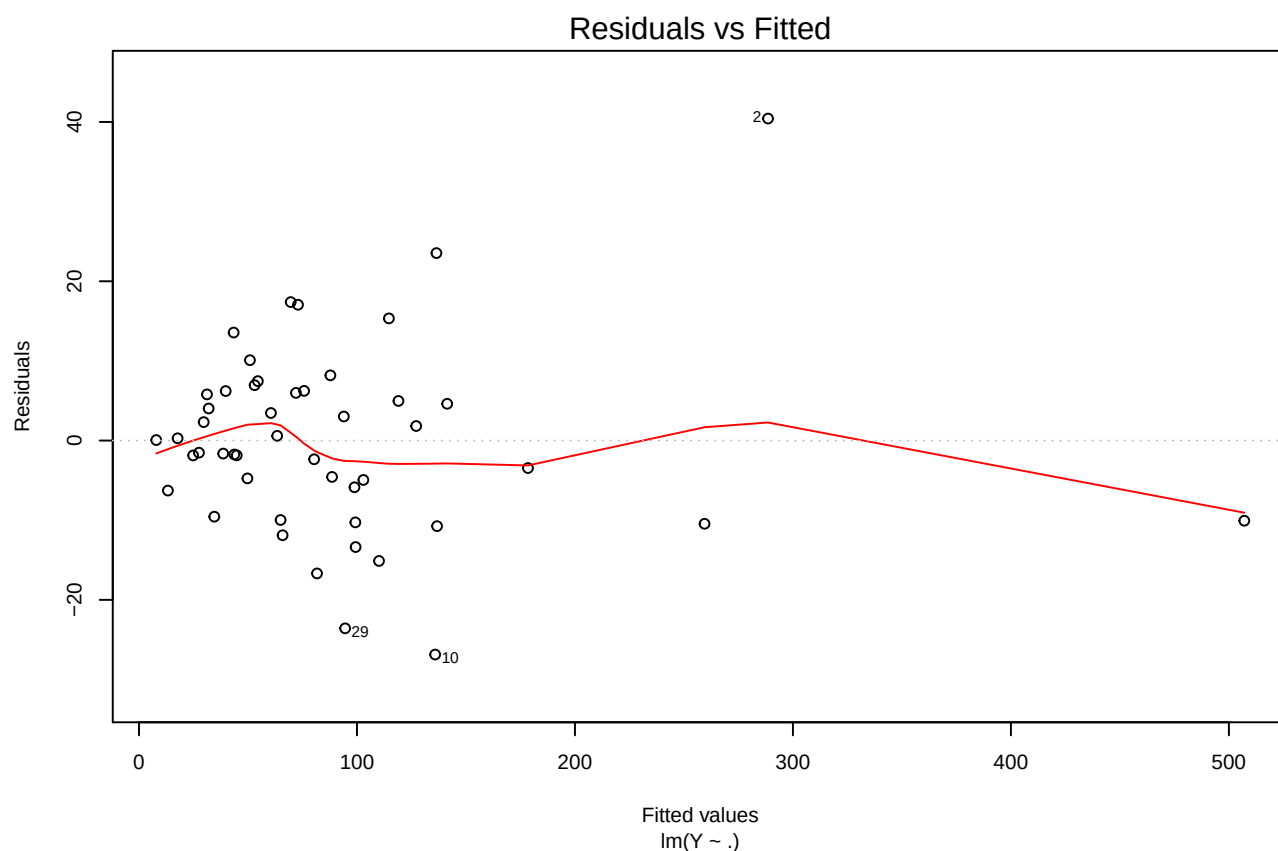
7.3.2 Plots of Raw Residuals against Predicted Value

We can plot the residuals, $\hat{e}_i$ against the predicted values, $\hat{Y}_i$, $i = 1, \dots, n$. The residuals are plotted on the vertical axis, while the predicted values are plotted on the horizontal axis. This plot may be used to look for trends in the data that may be indicative of a model **misfit**. If the regression assumptions are satisfied and the model fits, this plot should show a **random**

scatter of points and the spread of the residuals should be approximately the same over the whole range - i.e. the residual variance should be the same for all observations.

The residual vs predicted values plot may also indicate that the regression equation is missing a term. If the plot is clearly trending or contains a quadratic pattern, a linear or quadratic term should be included in the model.

```
fit1 <- lm(Y~., bankdf)
par(mfrow = c(1,1), mar = c(5,5,5,0), cex = 0.7)
plot(fit1, which=c(1))
```



The above figure plots the residuals of the full model of the Bank data against its fitted values. It indicates that the model is misspecified. The plot does not appear to be random and the variance of the residuals appear to be increasing as the size of the fitted values increases. This indicates that the residuals are not homoskedastic (i.e. they are heteroskedastic). The plot also indicates that some observations could be potential outliers. (all those observations with a "?" next to them) Notice also that the observations in the plot all roughly lie in the region $x = ay^2 + b$ for some a and b , where x = the fitted values and y = residuals. This indicates that the data might need to be transformed. Log transformations are often useful. In this case $\sqrt{Y}$ is also useful.

7.3.3 Plots of Residuals versus the Explanatory Variables

The residuals should be plotted against all of the explanatory variables. These plots should not show any pattern if the model fit is adequate. The plots can also be used in order to identify whether or not the explanatory variables should be transformed. If the plot is increasing like a fan, for say variable X_k , then $\log(X_k)$ or $\log(X_k + a)$, for some a , should be used. or if the plot is increasing like a parabolic fan, then $\sqrt{X_k}$ or $\sqrt{X_k + a}$, for some a , should be used.

7.4 Tests of Normality

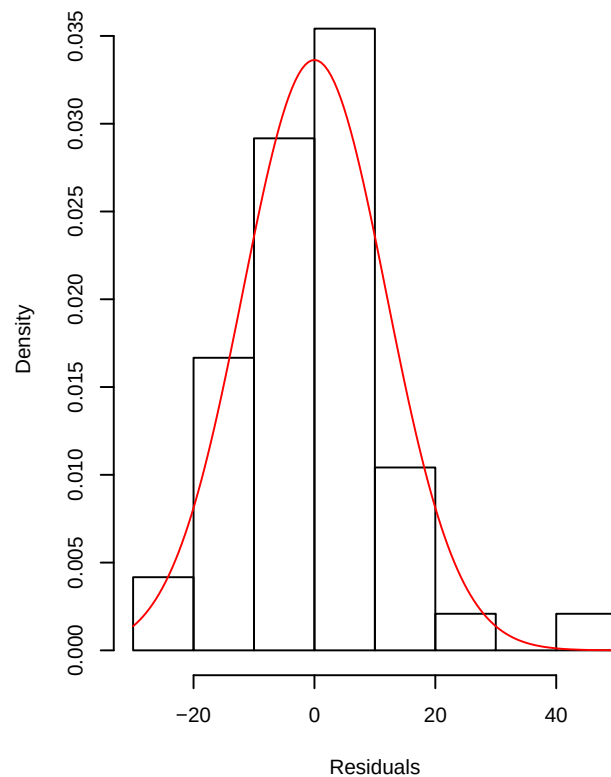
7.4.1 Normal Probability Plots

Most computer packages allows one to do a **normal probability** plot where the cumulative percent is plotted against the random variable - in our case the estimated residuals. If the error random variable is normally distributed the plot should be approximately a straight line as indicated in the graph below.

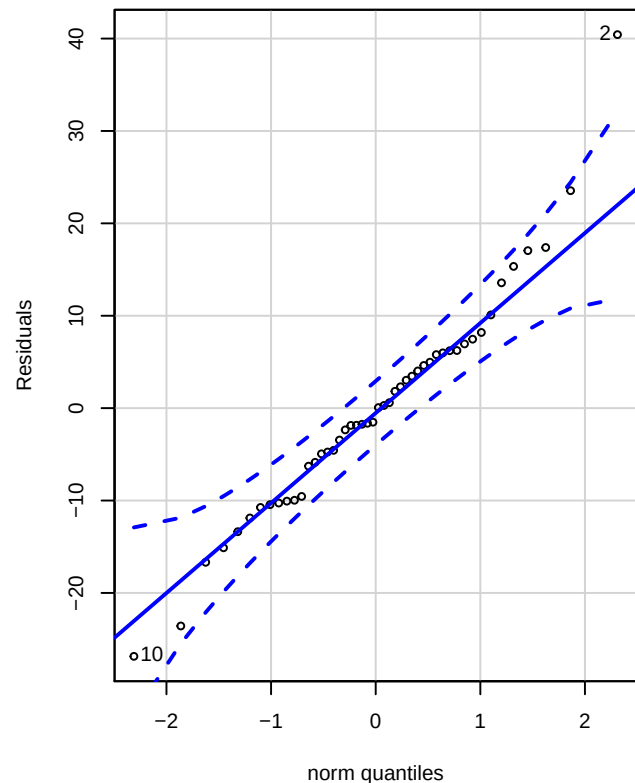
```
e      <- fit1$residuals
par(mfrow = c(1,2), mar = c(5,5,5,0), cex = 0.7)
hist(e, xlab = "Residuals",
main="Empirical distribution of the errors", prob=T)
m=mean(e); s=sd(e); rr=seq(-30,50,length.out=150)
lines(rr,dnorm(rr,m,s),col='red')

qqPlot(e, main = "QQ normal of Full Models Errors",
       ylab = "Residuals")
```

Empirical distribution of the errors



QQ normal of Full Models Errors



```
## [1] 2 10
```

7.4.2 Rankits - against the Residuals

Order the estimated residuals ($\hat{e}_{(1)} \leq \hat{e}_{(2)} \leq \dots \leq \hat{e}_{(n)}$) and plot them against the expected ordered $N(0,1)$ deviates. These expected ordered normal deviates are called *rankits* and denoted by $(z_1, \dots, z_n)$. This plot is often referred to as quantile-quantile plot (qq plot). The plot should be approximately a straight line. Departures from the straight line indicates that the normality assumption may be violated. There may also be outliers - points far away from the straight line at the end points.

The figures above displays the histogram and the qq plot of the estimated residuals. The histogram indicates that the residuals are skewed to the right with potential outliers. The qq plot is very close to a straight line and the assumption of normality is feasible although there may be long tails, indicating potential outliers.

7.4.3 Half-Normal Plots

If instead we plot the Rankits against the absolute value of the residual we get what is called a **half -normal** plot. This will highlight extreme values. It is also useful when the sample size is small.

7.4.4 Detrended Normal Plots

The **detrended normal** plot plots the deviation from the expected normal deviates against the residuals. The deviations are

$$\text{Deviations from expected} = \text{Expected normal value} - \frac{\hat{e}_j}{se(\hat{e}_j)}$$

The plot should show a straight line about zero.

7.5 Histograms

7.5.1 Histogram of Predicted values

The histogram of the predicted values, $\hat{Y}_j, j = 1, \dots, n$, for the Bank data can be plotted. This histogram indicates that $\hat{Y}$ is skew to the right. The skewness coefficient is 3.086. The kurtosis is high (14.956) indicating that the normality assumption may not be feasible.

7.5.2 Histogram of (Raw) Residuals

The histogram of the estimated residuals of the Bank data is displayed in figure ... above. The histogram indicates that the residual series is skewed to the right. The skewness coefficient is 0.607. The kurtosis is high (4.775) indicating that the normality assumption might not be valid.

7.6 Formal Statistical Tests for Normality

There are several formal tests for normality, namely, the χ^2 goodness of fit test, the Kolmogorov-Smirnov Test and the Shapiro-Wilk's test and the Jarque Bera test.

7.6.1 Kolmogorov-Smirnov Test

Let $(X_1, \dots, X_n)$ be a random sample from $\mathbf{X}$ with distribution function $F(x)$. Let $\bar{F}_n(x)$ be the cumulative sample distribution function, i.e.

$$\bar{F}_n(x) = \frac{\text{number of } X_i \leq x}{n}$$

The Kolmogorov-Smirnov statistic for testing the Hypotheses: $H_0 : F = F_0$ vs $H_1 : F \neq F_0$ is defined by

$$D_n = \sup_{-\infty < x < \infty} |\bar{F}_n(x) - F_0(x)|$$

Reject H_0 in favour of H_1 if D_n is larger than the significance value. The cutoff values are tabulated in Miller (1956).

In the normal case with μ and σ^2 unknown the Kolmogorov-Smirnov Test statistic is

$$D_n^* = \sup_{-\infty < x < \infty} \left| \bar{F}_n(x) - \Phi\left(\left(\frac{x - \bar{X}}{s}\right)\right) \right|$$

Critical values of D_n^* have been computed by Liliefors(1967).

```
ks.test(fit1$res, "pnorm", mean(e), sd(e))

##
## One-sample Kolmogorov-Smirnov test
##
## data: fit1$res
## D = 0.099125, p-value = 0.696
## alternative hypothesis: two-sided
```

Using R we see that the normality assumption cannot be rejected for the residuals of full model of the Bank data set.

7.6.2 The Shapiro-Wilk Test

A more powerful test for normality is given by Shapiro-Wilk. Let $(X_1, \dots, X_n)$ be a random sample from $N(\mu, \sigma^2)$ and let $(X_{(1)}, \dots, X_{(n)})$ be the order statistics. Consider the correlation R^2

between $X_{(i)}$, and $\Phi^{-1}(\frac{i-1/2}{n})$

$$R^2 = \frac{\left(\sum_{i=1}^n (X_{(i)} - \bar{X}) \Phi^{-1}(\frac{i-1/2}{n})\right)^2}{\sum_{i=1}^n (X_{(i)} - \bar{X})^2 \sum_{i=1}^n \Phi^{-1}(\frac{i-1/2}{n})^2}$$

If a variable is normally distributed, the R^2 value is close to one while if it is not normally distributed, the R^2 value will be small.

R is used to perform the Shapiro Wilks test on the residuals of the full model of the Bank Data. The test indicates that the normality assumption cannot be rejected.

```
shapiro.test(fit1$res)

##
##  Shapiro-Wilk normality test
##
## data:  fit1$res
## W = 0.96371, p-value = 0.1426
```

7.6.3 Skewness and Kurtosis as Tests for Normality

If $(X_1, \dots, X_n)$ is a sample from $N(\mu, \sigma^2)$ the sample skewness coefficient is

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})^3}{ns^3}$$

and the kurtosis is

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^4}{ns^4}$$

For large samples

$$b_1 \approx N\left(0, \frac{6}{n}\right) \text{ or } Z_1 = \frac{b_1}{\sqrt{6/n}} \approx N(0, 1)$$

and

$$b_2 - 3 \approx N\left(0, \frac{24}{n}\right) \text{ or } Z_2 = \frac{b_2 - 3}{\sqrt{24/n}} \approx N(0, 1)$$

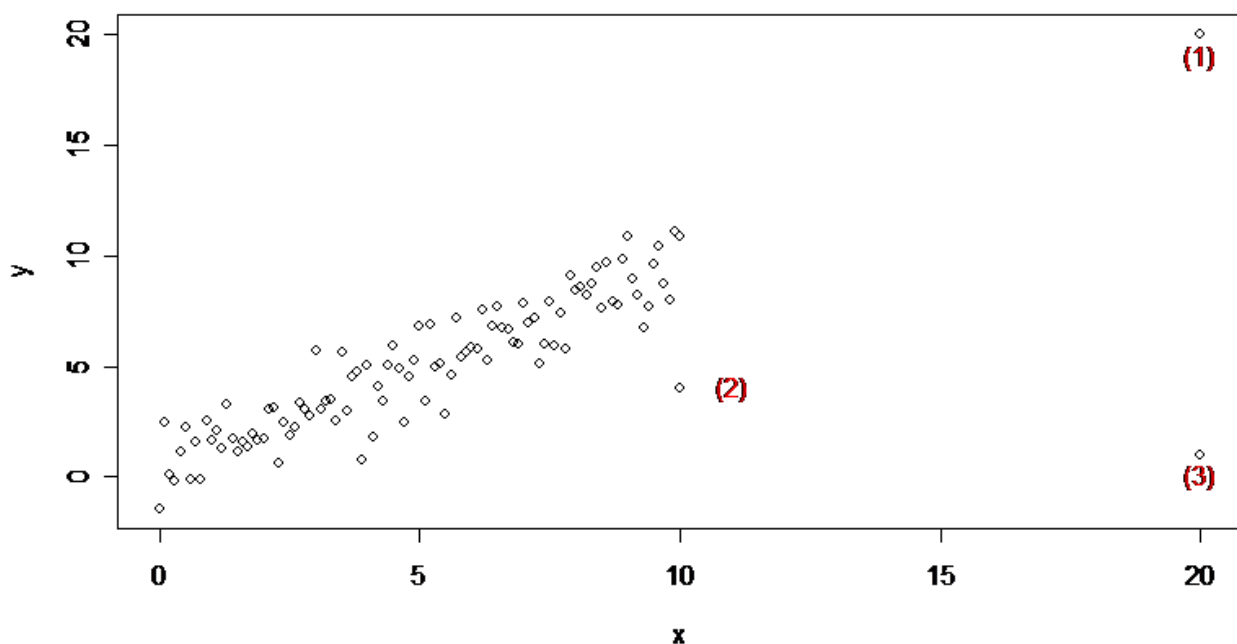
An Jarque Bera test combines Z_1 and Z_2 and can be used to test for normality

$$Z_1^2 + Z_2^2 \sim \chi_2^2$$

7.7 Detection of Outliers and Influential Points

When plotting residuals against predicted values the plot may indicate **outliers**, or an **influential points**, or both. (Both will be defined later.)

The figure below displays outliers and influential observations. Observation (1) is an outlier in the X space, observation (2) is potentially influential while observation (3) is both outlying and influential. As an example, consider all data points excluding points 1, 2 and 3. The estimated line will be roughly $y = x$. When observation 3 is now included into the model, the fitted line will run through point 3 and the slope will be approximately equal to 0. The **estimated parameters are significantly altered** indicating that point 3 is an influential observation.



Observations with a **residual close to zero** or **exactly zero** should be investigated since they may be influential observations. Influential observations should be deleted from the data set and the model should be refit. If the observations are truly influential observations, the estimated parameters will normally deviate significantly from the initial fitted model.

The problem becomes very complicated when there are several explanatory variables in the regression model, and not only one explanatory variable, as indicated above. The plot of the residuals with the predicted values may not show up anything and more advanced techniques in multivariate analysis may then be useful.

7.8 The analysis of Residuals

7.8.1 Deleted Observations, Outliers and Studentized Residuals

In this section we investigate various statistics that can be used in order to test whether a particular observation is an outlier or an influential observation. Large residuals relative to the standard deviation of the residuals should be investigated. The variance of the i^{th} observation is

$$var(\hat{e}_i) = \sigma^2(1 - h_{ii})$$

, where h_{ii} is the i^{th} diagonal element of the Hat matrix (see 19). If σ^2 were known, we could use

$$z_i = \frac{\hat{e}_i}{\sqrt{\sigma^2(1 - h_{ii})}} \sim N(0, 1)$$

in order to identify outlying observations. Since σ^2 is unknown, s^2 can be used as an estimate of σ^2 such that

$$z_i = \frac{\hat{e}_i}{\sqrt{s^2(1 - h_{ii})}} \quad (20)$$

z_i is known as the **standardized residuals**. It does not follow a normal distribution but values greater than 2 in absolute should be considered as potential outliers.

One way of determining the effect or *influence* of an observation is to delete the observation and then redo all calculations.

Let $\hat{\beta}_{(i)}$ be the least squares estimate of β with i^{th} observation deleted. Let $\mathbf{X}_{(i)}$, $\mathbf{Y}_{(i)}$ and $s_{(i)}^2$ be similarly defined then

$$\hat{\beta}_{(i)} = (\mathbf{X}_{(i)}' \mathbf{X}_{(i)})^{-1} \mathbf{X}_{(i)}' \mathbf{Y}_{(i)} \quad (21)$$

and

$$s_{(i)}^2 = \frac{1}{n - k - 1} \sum_{i=1}^{n-1} (Y_i - \mathbf{x}_i \hat{\beta}_{(i)})^2 \quad (22)$$

with $\mathbf{x}_i$ the i^{th} row of $\mathbf{X}$. Then

$$\hat{Y}_{(i)} = \mathbf{x}_i \hat{\beta}_{(i)}$$

is the prediction at $\mathbf{x}_i$ with observation i deleted. If

$$u_i = Y_i - \hat{Y}_{(i)}$$

then $E(u_i) = 0$ and the variance of u_i is

$$\text{var}(u_i) = \sigma^2(1 + \mathbf{x}_i(\mathbf{X}_{(i)}'\mathbf{X}_{(i)})^{-1}\mathbf{x}_i') \quad (23)$$

If σ^2 is estimated by $s_{(i)}^2$ which is independent of Y_i then the statistic

$$t_i = \frac{Y_i - \mathbf{x}_i\hat{\beta}_{(i)}}{\sqrt{s_{(i)}^2(1 + \mathbf{x}_i(\mathbf{X}_{(i)}'\mathbf{X}_{(i)})^{-1}\mathbf{x}_i')}} \quad (24)$$

will have a *Student's t* distribution with $n - k - 1$ degrees of freedom. Standard matrix results can be used to show that this t_i reduces to

$$t_i = \frac{\hat{e}_i}{\sqrt{s_{(i)}^2(1 - h_{ii})}} \quad (25)$$

where the $s_{(i)}^2$ given by equation (22) can be calculated as

$$s_{(i)}^2 = \frac{1}{n - k - 1} \left(\hat{e}'\hat{e} - \frac{\hat{e}_i^2}{1 - h_{ii}} \right) \quad (26)$$

t_i is known as the **studentized residual** and potential outliers have $|t_i| \geq t_{n-k-1}^{(\alpha/2)}$. It is also *strongly recommended* that the studentized residuals instead of the residuals be used in residual plots of the kind discussed before. This is because cut off lines can be drawn at $\pm t_{n-k-1}^{(\alpha/2)}$ (or ± 2 if n is large) and observations outside these rough lines can be investigated further.

We can show that

$$t_i = z_i \sqrt{\frac{n - k - 1}{n - k - z_i^2}}$$

7.8.2 Measures of Influence: Leverage

Outliers can cause harmful effects on OLS estimation but they are not as serious as influential observations. Influential observations often have high leverage (h_{ii}). It has been recommended that points with $h_{ii} \geq \frac{2k}{n}$ should be considered as influential observations. This can be seen by calculating the average value of h_{ii} .

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

, but $\mathbf{H}$ is idempotent (such that $\mathbf{H}^2 = \mathbf{H}$, and $\mathbf{H} = \mathbf{H}'$). Then

$$tr(\mathbf{H}) = tr(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = tr((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) = tr(\mathbf{I}_k) = k$$

But $tr(\mathbf{H}) = \sum_{i=1}^n h_{ii} = k$ such that the average value for h_{ii} , is $\frac{k}{n}$.

From $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$, any observation can be written as

$$\hat{Y}_i = h_{i1}Y_1 + h_{i2}Y_2 + \cdots + h_{ii}Y_i + \cdots + h_{in}Y_n \quad (27)$$

Let us consider the case where $n = 2$. Then

$$\mathbf{H} = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix} \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix} = \begin{pmatrix} h_{11}^2 + h_{12}^2 & \cdots \\ \cdots & h_{12}^2 + h_{22}^2 \end{pmatrix}$$

thus $h_{11} = h_{11}^2 + h_{12}^2$ and $h_{22} = h_{12}^2 + h_{22}^2$. In general

$$h_{ii} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i}^n h_{ij}^2$$

such that $0 \leq h_{ii} \leq 1$. If $h_{ii} = 0$, then all $h_{ij} = 0$, while if $h_{ii} = 1$, then all $h_{ij} = 0$, for $j \neq i$.

From (27), if $h_{ii} = 1$, then $\hat{Y}_i = Y_i$, and $\hat{e}_i = 0$ implying that the fitted line will run through the point (Y_i, X_i) (only one explanatory variable). We can show that h_{ii} is the distance from the point $(X_{i1}, X_{i2}, \dots, X_{ip})$ to the centre of the X -data. But $0 \leq h_{ii} \leq 1$. So large values of h_{ii} , - close to 1, should be of concern. The suggested cut-off value is twice the average value, $\frac{2k}{n}$. Any points above the cut-off point must be carefully investigated.

7.8.3 Measures of Influence: Outliers and Influential Observations

Cook has developed a statistic to detect outliers and influential observations at the same time. Cook's statistic is based on

$$F = \frac{(\hat{\beta} - \beta)'(\mathbf{C})^{-1}(\hat{\beta} - \beta)}{ks^2} = \frac{(\hat{\beta} - \beta)'\mathbf{X}'\mathbf{X}(\hat{\beta} - \beta)}{ks^2} \sim F_{k, n-k}$$

where $\mathbf{C}^{-1} = (\mathbf{X}'\mathbf{X})$.

Cook replaced the population parameter β with the estimate $\hat{\beta}$ and the estimate $\hat{\beta}$ is replaced by

$$\hat{\beta}_{(i)} = (\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{X}'_{(i)}\mathbf{Y}_{(i)}$$

Cook's statistic is then

$$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})'\mathbf{X}'\mathbf{X}(\hat{\beta}_{(i)} - \hat{\beta})}{ks^2} \quad (28)$$

Since $\hat{\beta}_{(i)}$ and $\hat{\beta}$ are dependent random variables the statistic D_i does not have a F - distribution, but Cook argued that it may well behave like a F -distribution. More importantly, however, Cook argued that if the deletion of the i^{th} observation has no effect on the estimates, then $\hat{\beta}_{(i)}$ and $\hat{\beta}$ will be close and D_i will be close to zero. On the other hand if after deleting the i^{th} observation $\hat{\beta}_{(i)}$ and $\hat{\beta}$ are substantially different, and D_i is large, then the i^{th} observation should be considered influential and should be investigated further.

A tentative cut off value of $\frac{4}{n-k-1}$ has been suggested in the literature.

Troskie prefers using

$$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})'\mathbf{X}'\mathbf{X}(\hat{\beta}_{(i)} - \hat{\beta})}{ks_{(i)}^2} \quad (29)$$

In all future situations we will use (29) when referring to Cook's distance.

Using matrix algebra it can be shown that

$$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})'\mathbf{X}'\mathbf{X}(\hat{\beta}_{(i)} - \hat{\beta})}{ks_{(i)}^2} = \frac{t_i^2}{k} \frac{h_{ii}}{1 - h_{ii}} \quad (30)$$

indicating that D_i combines the outlying effect of the i^{th} observation (as measured by t_i) and the leverage (as measured by h_{ii}).

A modification of Cook's distance was proposed by Atkinson (called Modified Cook). It is defined as

$$A_i = \text{Mod Cook} = \sqrt{n-k}|t_i| \sqrt{\frac{h_{ii}}{1-h_{ii}}} = \sqrt{n-k} \sqrt{D_i} \quad (31)$$

The proposed cut-off values are either 2 or 3.

Example 10: Residual plots of the Bank Data

Function for plotting:

```
OUTLIERS<-function(XDATA,YDATA){
  XDATA<-as.matrix(XDATA); YDATA<-as.matrix(YDATA)
  p<-NCOL(XDATA); n<-NROW(XDATA); k<-1+p

  # the x and y matrix used in the anlaysis
  x<-matrix(c(rep(1,n),XDATA),nrow=n,ncol=k)
  y<-matrix(YDATA)

  # the beta matrix and fitted
  BETA<-solve(t(x)%*%x)%*%t(x)%*%y
  fitted<-x%*%BETA

  # the residuals
  ei<-y-x%*%BETA
  # the residual variance
  e.e<-t(ei)%*%ei
  s2<-e.e[1]/(n-k)

  # the hat matrix
  H= x%*%solve(t(x)%*%x)%*%t(x)
  # the diagonal elements of the hat matrix
  hii<-diag(H)
  # the standardised residuals
  zi<-ei/sqrt(s2*(1-hii))
  # the studentised residuals
  s2.i<-(e.e[1]-(ei^2)/(1-hii))/(n-k-1)
  ti<-ei/sqrt(s2.i*(1-hii))
  # Cooks distance
  Di<-(zi^2)*hii/(k*(1-hii))
  # Cooks distance as preferred by Troskie
  di<-(ti^2)*hii/(k*(1-hii))
  # Modified Cooks statistic : Atkinsons Statistic
  Ai<-sqrt((n-k)*di)

  #the diagnostic plots
  par(mfrow=c(3,2),mar=c(4,4,3,3))
  plot(ti,type="n",main="Studentised Residuals",ylab="Studentised Residuals")
  text(ti,labels=as.character(1:n),cex=0.85)
  abline(h=c(-2,2),lty=2,col="red",lwd=2); abline(h=0,lty=2,col="black")

  plot(hii,type="n",main="The Leverage of the i'th Observation",ylab="Leverage")
  text(hii,labels=as.character(1:n),cex=0.85)
  abline(h=2*k/n,lty=2,col="red",lwd=2); abline(h=0,lty=2,col="black")

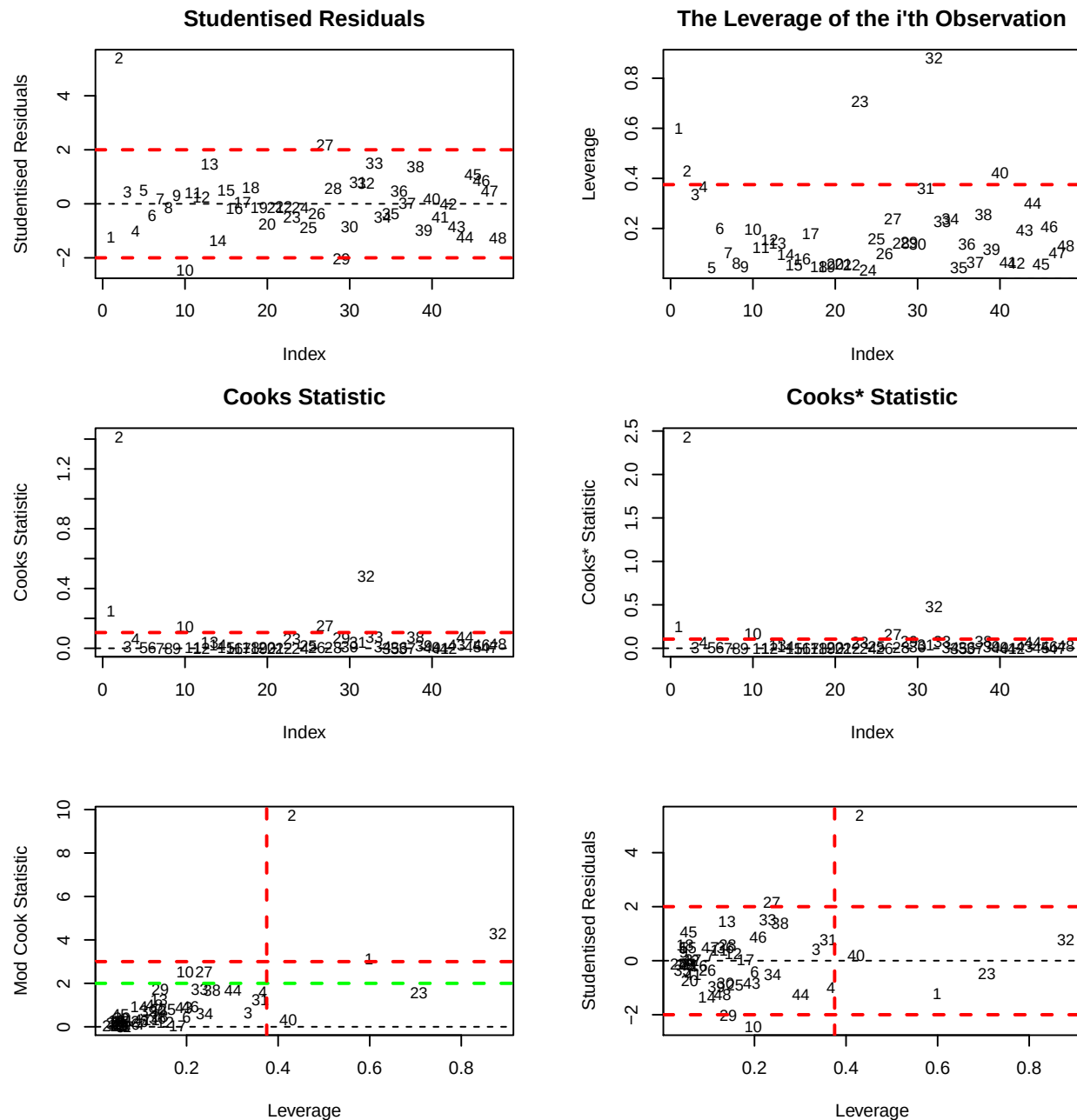
  plot(Di,type="n",main="Cooks Statistic",ylab="Cooks Statistic")
  text(Di,labels=as.character(1:n),cex=0.85)
  abline(h=4/(n-k-1),lty=2,col="red",lwd=2); abline(h=0,lty=2,col="black")

  plot(di,type="n",main="Cooks* Statistic",ylab="Cooks* Statistic")
  text(di,labels=as.character(1:n),cex=0.85)
  abline(h=4/(n-k-1),lty=2,col="red",lwd=2); abline(h=0,lty=2,col="black")

  plot(hii,Ai,type="n",xlab="Leverage",ylab="Mod Cook Statistic")
  text(hii,Ai,labels=as.character(1:n),cex=0.85)
  abline(h=c(-3,3),lty=2,col="red",lwd=2)
  abline(h=c(-2,2),lty=2,col="green",lwd=2)
  abline(h=0,lty=2,col="black"); abline(v=0,lty=2,col="black")
  abline(v=2*k/n,lty=2,col="red",lwd=2)

  plot(hii,ti,type="n",xlab="Leverage",ylab="Studentised Residuals")
  text(hii,ti,labels=as.character(1:n),cex=0.85)
  abline(h=c(-2,2),lty=2,col="red",lwd=2)
  abline(h=0,lty=2,col="black"); abline(v=0,lty=2,col="black")
  abline(v=2*k/n,lty=2,col="red",lwd=2)
}
```

```
Xdf <- bankdf[, c(2:9)]; Ydf <- bankdf[, c(1)]
OUTLIERS(Xdf, Ydf)
```



The figure above can be used in order to identify all potential outliers and influential observations in the Bank Data set when fitting the full model. The plot displays the studentised residuals, the leverage, Cook's statistic, Modified Cook's statistic and their respective cut off values. The figures indicate that observation 2 is a potential outlier while observations 1, 23 and 32 are potential influential observations. It is recommended that these observations should be removed and the model should then be estimated again.

8 Variable Selection Procedures

Model building entails selecting those variables that are deemed important to the area under investigation. In this section it is assumed that variable selection and model selection are equivalent processes. Rawlings *et al.* (1998) stresses that the elimination of variables from the model is dependent on the aims of the study. It is stressed that variable selection procedures are relatively unimportant if the researcher's aim is to provide a simple description of the behaviour of the response variable in a particular data set. Draper and Smith (1966) adds that variable selection should be undertaken so as to provide a linear model that is *"useful for prediction purposes and includes as many variables as possible so as to provide adequate fitted values for a data set."* It is however stressed that researchers should consider the cost of acquiring information about the variables to be included in the final model. In general variable selection entails making a compromise between the last two points since the monitoring of many variables may be too expensive. Miller (1990) notes the importance of finding a small subset of variables that provides adequate fit and precision.

The following regression variable selection techniques are the most popular:

(1) All Possible Regressions, (2) Stepwise Procedures and (3) Information criteria such as AIC and BIC.

8.1 All Subsets Regression

Consider the linear regression model: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + e$. There exists $2^p - 1$ different regression equation equations that can be undertaken with the p explanatory variables. The **all subsets regression** procedure entails fitting all $2^p - 1$ equations and then judging which regression equation or group of regression equations best fit the data set based on some criterion (R^2 , R_{adj}^2 , MSE or C_p). Often researchers fit all one variable models and then choose the best of of these models. All two variable models are then fitted in order to select the best two variable model. This process is then continued for all subset sizes until p potential models have been found. At this stage the researcher then compares the different regression equations (based on various criterion) in order to put forward one or more competing regression equations.

8.1.1 The R^2 Criterion

The coefficient of multiple determination, R^2 , defined as

$$R^2 = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2}$$

measures the amount of variation explained by the linear regression. When used for model selection, the aim is to select a model that maximises the R^2 statistic. Strict application of this criterion would ensure that the maximum R^2 model would contain all explanatory variables since the statistic cannot decrease with the inclusion of new variables into the regression equation. A visual graph of R^2 against the number of variables considered might be appropriate in order to judge the marginal increase in R^2 by the addition of new variables into the regression equation. The final model selected under this criterion would thus be the model for which R^2 has stabilised close to its maximum.

8.1.2 The Adjusted R^2 Criterion

The adjusted R^2 statistic, R_{adj}^2 , defined as

$$R_{adj}^2 = 1 - \frac{n-1}{n-p} (1 - R^2)$$

takes account of the number of explanatory variables included in the regression model. R_{adj}^2 does not need to increase as the number of variables increase since the increase in the R^2 statistic is adjusted by the increase in the number of variables in the new regression equation. As new variables enter into the regression equation, R_{adj}^2 tends to stabilise. Rawlings *et al.* (1998) states that the simplest model with R_{adj}^2 near to this stabilised value should be chosen.

8.1.3 The Residual Mean Square Criterion

The residual mean square (MSE) is often used as an estimate of the residual variance, σ^2 . Draper and Smith (1966) show that σ^2 is expected to decrease as more important variables enter into the regression equation such that MSE will tend to stabilise as the number of variables included in the equation becomes large. In many applications the chosen model is the one that minimises the MSE.

8.1.4 Mallows C_p Criterion

The Mallows C_p criterion (Mallows (1964)) is defined as

$$C_p = \frac{p \sum \hat{e}_i^2}{s^2} + 2p - n$$

where $\sum \hat{e}_i^2$ is the residual sums of squares from the p variable model, s^2 is the estimate of σ^2 based on all of the explanatory variables and n is the sample size.

Assuming that a p parameter model is appropriate then $E(C_p) = p$. It follows that the plot of C_p versus p will indicate potentially adequate models. Such models will be close to the $C_p = p$ line. Draper and Smith (1981) suggests that models with low C_p with a value close to p should be preferred to models with higher C_p values.

8.1.5 AIC and BIC

In a regression context AIC and BIC can be rewritten as

$$\begin{aligned} AIC &= n \ln \left(\sum \hat{e}_i^2 \right) + 2p - n \ln(n) \\ BIC &= n \ln \left(\sum \hat{e}_i^2 \right) + p \ln(n) - n \ln(n) \end{aligned}$$

which could now be used as a model selection procedure. The appropriate model selected is the model that minimises the AIC or the BIC measure.

It should be noted that all of the above procedures should be used as a guide in the selection of an appropriate regression model. Rawlings *et al.* (1998) states that "*no variable selection procedure can substitute for the insight of the researcher.*" With reference to the C_p , Mallows (1973) comments that " *C_p cannot be expected to provide a single best equation.*" Draper and Smith (1981) agrees with the above statement and adds, "*Nor can any other selection procedure. All selection procedures are essentially methods for the orderly displaying and reviewing of the data. applied with common sense, they can provide useful results; applied thoughtlessly, and/or mechanically, they may be useless or even misleading.*"

8.2 Stepwise Regression Procedures

Stepwise procedures might be preferred to the all subsets procedure due to the amount of computation required in fitting all possible regression models. Stepwise procedures use partial F tests in order to investigate whether or not a variable should be added or deleted from

a regression equation. These techniques require the user to specify two F statistics called F-to-enter (or F_{in}) and F-to-leave (or F_{out}). F_{in} is usually set equal to a value between 1 and 4 while F_{out} is often set equal to a value slightly smaller than F_{in} .

8.2.1 Backward Selection

The Backward elimination procedure starts off by fitting the regression equation containing all of the variables considered and then searches for which variables to eliminate from the regression equation. The procedure consists of the following four steps:

1. Decide upon a value for F_{out} .
2. Fit the regression equation containing all of the variables.
3. Calculate the partial F-test value for each variable as though it were the last variable to enter the regression equation. Note that the partial F-test value for each variable is equal to the square of the t-statistics of the beta coefficients such that the F-test value for the i^{th} variable is equal to

$$F_i = \frac{\hat{\beta}_i^2}{v_{ii}} = t_i^2 \sim F_{1,n-p-1}$$

where v_{ii} are the diagonal elements of the variance covariance matrix of the beta coefficients. (i.e. $s^2(X'X)^{-1}$)

4. Now compare the smallest F_i values with F_{out} .
 - (a) If this value is smaller than F_{out} the associated variable is deleted from the regression equation and the process is repeated by considering only the remaining variables.
 - (b) If the value is greater than F_{out} the process is stopped and the final model has been found.

Draper and Smith (1966) suggest that the above procedure can provide satisfactory results although cautions to the use of the procedure if the X matrix is ill conditioned. In this regard Troskie (1999) notes that *"Such collinearities can have disastrous effects on the OLS and MLE estimates. It is well known, that because of collinearities, that the backward procedure can give entirely different results from the forward selection procedure."* Forward Selection is discussed next.

8.2.2 Forward Selection

The Forward Selection procedure is the opposite of the Backward elimination procedure. The procedure starts off by including the variable that exhibits the highest correlation with the response variable (Y) and then searches for which variables to include in the regression equation by examining the F-test values of the variables not already in the equation. The procedure consists of the following three steps:

1. Decide upon a value for F_{in} . As stated above F_{in} is usually set equal to a value between 1 and 4, since the value 4 corresponds with a t-statistic value of 2.
2. Determine which variable is most correlated with the response variable, say X_1 then fit the regression model: $Y = \beta_0 + \beta_1 X_1 + e$. If this regression is not significant the procedure stops and the response variable can only be modelled by its mean, $\bar{Y}$.
3. Fit all two variable models containing X_1 and then calculate the partial F-test value for each variable to enter the regression equation given that X_1 is already in the model. (Once again this is simply equal to the square of the t-statistic of the beta coefficient of the new variable that enters the equation.)
 - (a) If the variable with the largest F_i value is greater than F_{in} , the variable is included into the regression equation and the process is continued by considering all three-, four-, five, ... p -variable equations containing the previous two-, three-, four-, ... $p - 1$ -variables respectively. The procedure is continued by adding a new variable to the regression equation until
 - (b) The largest F_i is smaller than F_{in} .

8.2.3 Stepwise Regression

Note that when undertaking both forward- and backward selection that variables enter and leave the model one at a time. Once a variable has entered into the regression equation they may not be deleted. Similarly in the backward elimination procedure once a variable has been deleted from the equation it cannot re-enter the model. Both procedures do not consider the effect that the inclusion or deletion of a variable has on the other variables in the model. In this regard it should be noted that a variable added early on in the procedure might become insignificant when other variables enter the equation. Similarly, in the backward elimination procedure, a variable can become significant once a number of variables have left the model. Stepwise regression uses a combination of forward selection and backward elimination in order to solve the above problem.

Stepwise regression consists of the following steps:

1. Decide upon a value for F_{in} and F_{out} .
2. Determine which variable is most correlated with the response variable, say X_1 then fit the regression model: $Y = \beta_0 + \beta_1 X_1 + e$. If this regression is not significant the procedure stops and the response variable can only be modelled by its mean, $\bar{Y}$.
3. Fit all two variable models containing X_1 and then calculate the partial F-test value for each variable to enter the regression equation given that X_1 is already in the model.
 - (a) If the variable, say X_2 , with the largest F_i value is greater than F_{in} , the variable is included into the regression equation.
 - (b) If the variable with the largest F_i value is smaller than F_{in} , the procedure stops.
4. Fit all three variable models containing X_1 and X_2 and then calculate the partial F-test value for each variable to enter the regression equation given that X_1 and X_2 is already in the model.
 - (a) If the variable, say X_3 , with the largest F_i value is greater than F_{in} , the variable is included into the regression equation.
 - (b) If the variable with the smallest F_i value is smaller than F_{out} , the variable is deleted from the equation.
5. Step 4 is continued in this way by adding and deleting variables at each step until no more variables either enter or leave the regression equation.

9 The Gauss - Markoff Theorem

We have assumed that $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. If $E(\mathbf{e}) = \mathbf{0}$ and $E(\mathbf{e}\mathbf{e}') = \sigma^2 \mathbf{I}$ then we know that the least squares estimate for β is given by

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

The Gauss-Markoff theorem gives a statement of how good this estimate is.

Theorem: Gauss-Markoff

In the model $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ where $E(\mathbf{e}) = \mathbf{0}$ and $E(\mathbf{e}\mathbf{e}') = \sigma^2 \mathbf{I}_n$ the **OLS** estimate is **BLUE** (BEST LINEAR UNBIASED ESTIMATE) of β . ■

Proof

Let $\beta^* = \mathbf{A}\mathbf{Y}$ for any matrix $\mathbf{A}$ (this is linear). Choose $\mathbf{A}$ to be

$$\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}$$

where $\mathbf{B}$ is arbitrary. To be unbiased

$$\begin{aligned} E(\beta^*) &= E(\mathbf{A}\mathbf{Y}) = ((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B})\mathbf{Y} \\ &= E((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B})(\mathbf{X}\beta + \mathbf{e}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + \mathbf{B}\mathbf{X}\beta = \beta \text{ (to be unbiased)} \\ &= \beta + \mathbf{B}\mathbf{X}\beta = \beta \end{aligned}$$

Thus $\mathbf{B}\mathbf{X} = \mathbf{0}$ to be unbiased.

The covariance matrix of β^* is

$$\begin{aligned} cov(\beta^*) &= E(\beta^* - \beta)(\beta^* - \beta)' \\ &= E[\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}\mathbf{Y} - \beta][\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}\mathbf{Y} - \beta]' \\ &= E[\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}(\mathbf{X}\beta + \mathbf{e}) - \beta][\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}(\mathbf{X}\beta + \mathbf{e}) - \beta]' \\ &= E[\{\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} + \mathbf{B}\mathbf{e} - \beta\}[\{\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} + \mathbf{B}\mathbf{e} - \beta\}]' \text{ (BX = 0)} \\ &= E[\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} + \mathbf{B}\mathbf{e}\}[\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} + \mathbf{B}\mathbf{e}\}]' \\ &= E[\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}\mathbf{e}\mathbf{e}'\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}]' \\ &= [\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}E(\mathbf{e}\mathbf{e}')\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}]' \\ &= [\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}\sigma^2 \mathbf{I}\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}]' \\ &= \sigma^2\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\}' \\ &= \sigma^2\{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{B}' + \mathbf{B}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{B}\mathbf{B}'\} \end{aligned}$$

$$\begin{aligned} cov(\beta^*) &= \sigma^2\{(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{B}\mathbf{B}'\} \text{ since } \mathbf{B}\mathbf{X} = \mathbf{0} \text{ and } (\mathbf{B}\mathbf{X})' = \mathbf{X}'\mathbf{B}' = \mathbf{0} \\ &= \sigma^2\{(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{G}\} \text{ where } \mathbf{G} = \mathbf{B}\mathbf{B}' \end{aligned}$$

The variances $var(\beta_i^*)$ of $cov(\beta^*)$ are the diagonal elements of $cov(\beta^*)$. The best (minimum variance) estimates are those values for which the diagonal elements of

$$\sigma^2\{(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{G}\}$$

are a minimum. But $(\mathbf{X}'\mathbf{X})^{-1}$ is known fixed. Thus to minimize the diagonal elements of $cov(\beta^*)$ we must minimize the diagonal elements of $\mathbf{G}$ which is g_{ii} . But

$$\mathbf{G} = \mathbf{B}\mathbf{B}'$$

is positive semi-definite so that $g_{ii} \geq 0$. Thus the diagonal elements of $cov(\beta^*)$ will attain their minimum if $g_{ii} = 0$ for $i = 1, \dots, n$. But if $\mathbf{B} = (b_{ij})$ then $g_{ii} = \sum_{j=1}^n b_{ij}^2$. Therefore if $g_{ii} = 0$ for all i then it must be true that $b_{ij} = 0$ for all j and for all i . This implies that

$$\mathbf{B} = \mathbf{0}$$

which is compatible with the condition of unbiased $\mathbf{B}\mathbf{X} = \mathbf{0}$.

Thus $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}$ and $\beta^* = \hat{\beta}$ so that the OLS estimate is **BLUE**. ■

10 Transformations

Our general linear model is given by

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + e \quad (32)$$

Many other models can be adapted or transformed to the general linear model and the theory and applications will apply to the new transformed model.

Of special interest are polynomial models of the type

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \cdots + \beta_p X^p + e$$

Let $X_1 = X, X_2 = X^2, \dots, X_p = X^p$ and use model (32). Care must be taken that the degree p of the polynomial is not too high otherwise high powers of the type

$$\sum_{j=1}^n X_j^{2p} = \sum_{j=1}^n X_j^{2p}$$

could lead to very inaccurate results. If polynomials of high degrees are to be fitted rather use the method of orthogonal polynomials.

Other quadratic terms could also easily be fitted by using a transformation. Consider a model of the type

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_1 X_3 + \beta_4 \cos(X_4) + \beta_5 \log(X_5) + \cdots + \beta_p X_p + e$$

and using obvious transformations can be written in the form given by (32). As long as the new model can be transformed to a linear form of the type (32) then all our previous methods will apply.

There are many non-linear models that could be transformed to a linear form.

The multiplicative model is

$$Y = \alpha X_1^\beta X_2^\gamma X_3^\delta \varepsilon$$

and can be transformed to

$$\ln Y = \ln \alpha + \beta \ln X_1 + \gamma \ln X_2 + \delta \ln X_3 + \ln \varepsilon$$

The exponential model

$$Y = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p} \varepsilon$$

and can be transformed to

$$\ln Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \ln \varepsilon$$

The reciprocal model

$$Y = \frac{1}{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + e}$$

can be transformed to

$$\frac{1}{Y} = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + e.$$

The Gompertz model

$$Y = \frac{1}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + e}}$$

can be transformed to

$$\ln\left(\frac{1}{Y} - 1\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + e$$

The power transform models are very popular with

$$Y^* = \begin{cases} \frac{Y^\lambda - 1}{\lambda} & \text{for } \lambda \neq 0 \\ \ln Y & \text{for } \lambda = 0 \end{cases}$$

Note:

The transformations discussed in this chapter are but a few of the many currently being used to reduce complex models to linear ones. When, as we assume here, the predictor or explanatory variables are not subject to error, there are no problems in transforming them. However for transformations on the dependent or response variable, Y , one must check that the least squares assumptions are not violated by making the transformation. Often one can avoid transforming the response variable by searching for suitable transformations in the $\mathbf{X}$'s.

11 Indicator Variables

In this section we will briefly discuss how we can include indicator variables into regression analysis.

11.1 One independent qualitative variable

We often have qualitative variables in our data set which require special care when undertaking a regression analysis. e.g. Sex (Male or Female), Growth phase in the development of a company (Start, Middle, End) or Riskiness of a portfolio (No risk, Average risk or high risk)

Lets assume for the time being that we have collected the following data:

- Y = the monthly salary of an individual,
- X_1 = number of worked and
- $X_2 = \begin{cases} 1 & \text{if the person is a male} \\ 0 & \text{otherwise} \end{cases}$
- $X_3 = \begin{cases} 1 & \text{if the person is a female} \\ 0 & \text{otherwise} \end{cases}$

An example of some data could be the following:

Y	X_1	X_2	X_3
8000	16	0	1
5000	14	1	0
2000	12	1	0
1500	11	0	1
7000	14	1	0
4500	13	1	0
3000	12	1	0
2500	12	0	1
25000	18	0	1
12000	12	0	1

How would we now construct the regression line? Lets first try and attempt to estimate the beta coefficients using the standard formula, $\hat{\beta} = (X'X)^{-1} X'Y$. The X matrix would be

$$\begin{pmatrix} 1 & 16 & 0 & 1 \\ 1 & 14 & 1 & 0 \\ 1 & 12 & 1 & 0 \\ 1 & 11 & 0 & 1 \\ 1 & 14 & 1 & 0 \\ 1 & 13 & 1 & 0 \\ 1 & 12 & 1 & 0 \\ 1 & 12 & 0 & 1 \\ 1 & 18 & 0 & 1 \\ 1 & 12 & 0 & 1 \end{pmatrix}$$

and $(X'X)$ is equal to

$$\begin{pmatrix} 10 & 134 & 5 & 5 \\ 134 & 1838 & 65 & 69 \\ 5 & 65 & 5 & 0 \\ 5 & 69 & 0 & 5 \end{pmatrix}$$

Notice that the first column of $(X'X)$ is equal to the sum of the last two columns. $(X'X)^{-1}$ will thus not exist and we will not be able to calculate the beta estimates.

We do the following in order to solve this problem: We treat X_1 as per normal but we code X_2 as follows, $X_2^* = 0$ if the respondent is female and $X_2^* = 1$ if the respondent is male. This new variable is known as a FACTOR or indicator variable. We model the relationship between the explanatory variables as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2^* + e$$

Our new X matrix will now be

$$\begin{pmatrix} 1 & 16 & 0 \\ 1 & 14 & 1 \\ 1 & 12 & 1 \\ 1 & 11 & 0 \\ 1 & 14 & 1 \\ 1 & 13 & 1 \\ 1 & 12 & 1 \\ 1 & 12 & 0 \\ 1 & 18 & 0 \\ 1 & 12 & 0 \end{pmatrix}$$

such that the beta coefficients are now equal to

$$\begin{pmatrix} -23787 \\ 2433.8 \\ -3552.9 \end{pmatrix}$$

Notice that we now have a straight line model relating Y and X_1 conditional on the sex variable. The two models are as follows:

Males	$Y = (-23787 - 3552.9) + 2433.8X_1$	or	$Y = -27340. + 2433.8X_1$
Females	$Y = (-23787) + 2433.8X_1$	or	$Y = -23787 + 2433.8X_1$

In general the models are equal to

Males	$Y = \hat{\beta}_0 + \hat{\beta}_2 + \hat{\beta}_1 X_1$
Females	$Y = \hat{\beta}_0 + \hat{\beta}_1 X_1$

Notice that both equations have the same beta estimate for X_1 . The only difference is the intercept term. This intercept term represents how much higher/lower the response variable (monthly salary) is for males compared to females.

We could extend this example by including variables that have more than two FACTOR LEVELS, say f . i.e. we have variables that have more than two categories. In this case we would introduce $f - 1$ indicator variables.

As an example, let's include the variable STUDY into the analysis (and not use SEX). In this example STUDY represents the kind of studies undertaken by each individual after matriculation/A levels. The categories might be: Technikon, University, College, or No study. Let these groups be known as Group 1, ..., Group 4.

We would thus introduce three dummy variables

$$\begin{aligned} X_2^* &= \begin{cases} 1 & \text{if the person studied at Technikon} \\ 0 & \text{otherwise} \end{cases} \\ X_3^* &= \begin{cases} 1 & \text{if the person studied at University} \\ 0 & \text{otherwise} \end{cases} \\ X_4^* &= \begin{cases} 1 & \text{if the person studied at a College} \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

The resulting model would be

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2^* + \beta_3 X_3^* + \beta_4 X_4^* + e$$

or

Group 1	$Y = (\beta_0 + \beta_2) + \beta_1 X_1$
Group 2	$Y = (\beta_0 + \beta_3) + \beta_1 X_1$
Group 3	$Y = (\beta_0 + \beta_4) + \beta_1 X_1$
Group 4	$Y = \beta_0 + \beta_1 X_1$

Note: In statistical packages FACTORS are often treated differently. We would enter of the values of the categorical (Factor) variable as one variable. We then assign numerical values for each of the groups. e.g. Group 1 = 1, Group 2 = 2, Group 3 = 3 and Group 4 = 0.

Example 11: Salary Survey Data

The following example is taken from Chatterjee and Price (1977). "The objective of the survey was to identify and quantify those factors that determine salary differentials." (Chatterjee and Price (1977), page 75) The data set comprises of four variables namely: Annual Salary (*salary*) in Dollars, Experience (*exp*) measured in years, Education level (*educ*) and Management responsibility (*mgt*). The education and management variables are indicator variables. *educ* is coded 1 for the completion of high school, 2 for the completion of college and 3 for the completion of an advanced degree. *mgt* is coded 1 for a person with management responsibilities and 0 otherwise. These variables can be coded to factor variables directly in R, which is way of telling R to recognise each level of the variable as a distinct category.

```
salary.df      <- read.table("salary.txt", header = T)
salary.df$educ. <- factor(salary.df$educ)
salary.df$mgt.  <- factor(salary.df$mgt)
```

Chatterjee and Price (1977) assumed that there existed a linear relationship between salary and experience. The other two variables are then added to the regression model in order to identify the differences between combinations of education and management levels with reference to salary levels.

```
fit <- lm(salary ~ exp + educ. + mgt., data = salary.df)
summary(fit)

##
## Call:
```

```
## lm(formula = salary ~ exp + educ. + mgt., data = salary.df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1884.60  -653.60   22.23   844.85  1716.47
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8035.60     386.69  20.781 < 2e-16 ***
## exp           546.18      30.52  17.896 < 2e-16 ***
## educ.2       3144.04     361.97   8.686 7.73e-11 ***
## educ.3       2996.21     411.75   7.277 6.72e-09 ***
## mgt.1        6883.53     313.92  21.928 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1027 on 41 degrees of freedom
## Multiple R-squared:  0.9568, Adjusted R-squared:  0.9525
## F-statistic: 226.8 on 4 and 41 DF,  p-value: < 2.2e-16
```

Notice that the output now has two education variables and one management variable. This happens because we set *educ.* and *mgt.* as factors. Factors have different levels (or groups). Recall that *mgt* was coded as either 0 or 1. The first level of *mgt.* will be set equal to 0 (i.e. no management responsibilities) and the second level is set equal to 1. Similarly the first level of *educ.* is 1 (high school). the second level is college and the third level is advanced degree. The *fit1=lm(salary~exp+educ.+mgt.)* command undertakes the regression and treats the lowest levels of categorical variables as the "base line" case. i.e. we are actually estimating the following model

$$salary = \beta_0 + \beta_1 \text{exp} + \gamma_1 \text{college} + \gamma_2 \text{advanced} + \delta_1 \text{respons}$$

where

$$\begin{aligned} \text{college} &= \begin{cases} 1 & \text{if the person studied at college} \\ 0 & \text{otherwise} \end{cases} \\ \text{advanced} &= \begin{cases} 1 & \text{if the person completed an advanced degree} \\ 0 & \text{otherwise} \end{cases} \\ \text{respons} &= \begin{cases} 1 & \text{if the person has management responsibilities} \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

We thus have the following regression lines:

educ	mgt	Regression line		
high school	None	$Salary$	$= \beta_0$	$+ \beta_1 exp$
high school	Yes	$Salary$	$= \beta_0 + \delta_1$	$+ \beta_1 exp$
college	None	$Salary$	$= \beta_0 + \gamma_1$	$+ \beta_1 exp$
college	Yes	$Salary$	$= \beta_0 + \gamma_1 + \delta_1$	$+ \beta_1 exp$
advanced	None	$Salary$	$= \beta_0 + \gamma_2$	$+ \beta_1 exp$
advanced	Yes	$Salary$	$= \beta_0 + \gamma_2 + \delta_1$	$+ \beta_1 exp$

educ	mgt	Regression line		
high school	None	$Salary$	$= 8035.6$	$+546.18exp$
high school	Yes	$Salary$	$= 8035.6 + 6883.53$	$+546.18exp$
college	None	$Salary$	$= 8035.6 + 3144.04$	$+546.18exp$
college	Yes	$Salary$	$= 8035.6 + 3144.04 + 6883.53$	$+546.18exp$
advanced	None	$Salary$	$= 8035.6 + 2996.21$	$+546.18exp$
advanced	Yes	$Salary$	$= 8035.6 + 2996.21 + 6883.53$	$+546.18exp$

From the output we can see that the *exp* coefficient is equal to 546.18. This indicates that one additional years experience increases the annual salary by \$546.18. We are now in a position to compare the salary levels for the different education and management levels. If we compare people based on management responsibilities we can see that individuals with management responsibilities earn \$6883.53 more than those individuals without management responsibilities. The annual salary difference between individuals with a high school education level and college level is \$3144.04, the difference between individuals with a high school and individuals with an advanced degree is \$2996.21 and the difference between individuals with a college background and an advanced degree is $3144.04 - 2996.21 = 147.83$.

12 Theorems and Proofs

Theorem 2 If $\mathbf{X} \sim N(\mu, \Sigma)$, then the moment generating function of $\mathbf{X}$ is:

$$M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}}) = e^{\mathbf{t}'\mu + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t}}$$

Proof. The moment generating function of $\mathbf{X}$ is

$$M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}}) = E(e^{t_1 X_1 + \dots + t_p X_p})$$

Let $\mathbf{X} - \mu = \mathbf{C}\mathbf{Y}$, where $\mathbf{C}'\Sigma^{-1}\mathbf{C} = \mathbf{I}$, then

$$g(y_1, \dots, y_p) = \frac{1}{(2\pi)^{\frac{1}{2}p}} e^{-\frac{1}{2}\mathbf{y}'\mathbf{y}}$$

Let $\psi(\mathbf{u})$ be the mgf of $\mathbf{Y}$ i.e.

$$\begin{aligned} \psi(\mathbf{u}) &= E(e^{\mathbf{u}'\mathbf{Y}}) \\ &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} e^{\mathbf{u}'\mathbf{y}} \frac{1}{(2\pi)^{\frac{1}{2}p}} e^{-\frac{1}{2}\mathbf{y}'\mathbf{y}} dy_1 \dots dy_p \\ &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \frac{1}{(2\pi)^{\frac{1}{2}p}} e^{(u_1 y_1 + \dots + u_p y_p) - \frac{1}{2}(y_1^2 + y_2^2 + \dots + y_p^2)} dy_1, \dots, dy_p \\ &= \prod_{i=1}^p \frac{1}{(2\pi)^{\frac{1}{2}p}} \dots \int_{-\infty}^{\infty} e^{u_i y_i - \frac{1}{2}y_i^2} dy_i \\ &= \prod_{i=1}^p \frac{1}{(2\pi)^{(\frac{1}{2})}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y_i - u_i)^2 + \frac{1}{2}u_i^2} dy_i \\ &= \prod_{i=1}^p \frac{e^{\frac{1}{2}u_i^2}}{(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y_i - u_i)^2} dy_i \\ &= \prod_{i=1}^p e^{\frac{1}{2}u_i^2} \cdot 1 \\ &= e^{\frac{1}{2}\mathbf{u}'\mathbf{u}} \end{aligned}$$

But

$$\begin{aligned} \mathbf{X} - \mu &= \mathbf{C}\mathbf{Y} \\ M(\mathbf{t}) &= E(e^{\mathbf{t}'\mathbf{X}}) = E(e^{\mathbf{t}'(\mathbf{C}\mu + \mathbf{C}\mathbf{Y})}) = e^{\mathbf{t}'\mu} E(e^{\mathbf{t}'\mathbf{C}\mathbf{Y}}) = e^{\mathbf{t}'\mu} E(e^{(\mathbf{C}'\mathbf{t})'\mathbf{Y}}) \end{aligned}$$

Let $\mathbf{t}^* = \mathbf{C}'\mathbf{t}$, then

$$\begin{aligned} M(\mathbf{t}) &= e^{\mathbf{t}'\boldsymbol{\mu}} E(e^{\mathbf{t}^{*\prime}\mathbf{Y}}) \\ &= e^{\mathbf{t}'\boldsymbol{\mu}} e^{\frac{1}{2}\mathbf{t}^{*\prime}\boldsymbol{\Sigma}\mathbf{t}^*} \\ &= e^{\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}(\mathbf{C}'\mathbf{t})'(\mathbf{C}'\mathbf{t})} \\ &= e^{\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\mathbf{C}\mathbf{C}'\mathbf{t}} \end{aligned}$$

but

$$\begin{aligned} \mathbf{C}'\boldsymbol{\Sigma}^{-1}\mathbf{C} &= \mathbf{I} \\ (\mathbf{C}')^{-1}\mathbf{C}\boldsymbol{\Sigma}^{-1}\mathbf{C}\mathbf{C}^{-1} &= (\mathbf{C}')^{-1}\mathbf{I}\mathbf{C}^{-1} \\ \boldsymbol{\Sigma}^{-1} &= (\mathbf{C}')^{-1}\mathbf{C}^{-1} = (\mathbf{C}\mathbf{C}')^{-1} \end{aligned}$$

Thus

$$M(\mathbf{t}) = e^{\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t}}$$

■

Theorem 3 If $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then if $\mathbf{Y} = \mathbf{C}\mathbf{X}$ for any matrix/vector $\mathbf{C}$ then $\mathbf{Y} \sim N(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}')$.

Proof. The mgf of $\mathbf{X}$ is:

$$M_{\mathbf{X}}(\mathbf{t}) = e^{\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t}}$$

The mgf of $\mathbf{Y}$ is:

$$\begin{aligned} M_{\mathbf{Y}}(t) &= E(e^{\mathbf{t}'\mathbf{Y}}) = E(e^{\mathbf{t}'\mathbf{C}\mathbf{X}}) \\ &= E(e^{(\mathbf{C}'\mathbf{t})'\mathbf{X}}) = e^{(\mathbf{C}'\mathbf{t})'\boldsymbol{\mu} + \frac{1}{2}(\mathbf{C}'\mathbf{t})'\boldsymbol{\Sigma}(\mathbf{C}'\mathbf{t})} \\ &= e^{\mathbf{t}'\mathbf{C}\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\mathbf{C}\boldsymbol{\Sigma}\mathbf{C}'\mathbf{t}} \end{aligned}$$

This is the mgf of a multivariate normal with mean $\mathbf{C}\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}\boldsymbol{\Sigma}\mathbf{C}'$. Thus $\mathbf{Y} = \mathbf{C}\mathbf{X} \sim N(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}')$. ■

Theorem 4 Let $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{X}$ be partitioned into $\mathbf{X} = \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix}$, then $\mathbf{X}_{(q \times 1)}^{(1)} \sim N(\boldsymbol{\mu}^{(1)}, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_{(r \times 1)}^{(2)} \sim N(\boldsymbol{\mu}^{(2)}, \boldsymbol{\Sigma}_{22})$ where

$$\begin{aligned} \boldsymbol{\mu} &= \begin{pmatrix} \boldsymbol{\mu}^{(1)} \\ \boldsymbol{\mu}^{(2)} \end{pmatrix}_r \\ \boldsymbol{\Sigma} &= \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{21} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}_r \end{aligned}$$

Proof. Let

$$\begin{aligned}\mathbf{Y} &= \mathbf{CX} \\ \mathbf{C} &= (\mathbf{I}_q, \mathbf{O})\end{aligned}$$

then

$$\begin{aligned}\mathbf{Y} &= \mathbf{CX} = (\mathbf{I}_q, \mathbf{O}) \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix} = \mathbf{X}^{(1)} \\ \mathbf{C}\boldsymbol{\mu} &= (\mathbf{I}_q, \mathbf{O}) \begin{pmatrix} \boldsymbol{\mu}^{(1)} \\ \boldsymbol{\mu}^{(2)} \end{pmatrix} = \boldsymbol{\mu}^{(1)} \\ \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}' &= (\mathbf{I}_q, \mathbf{O}) \begin{pmatrix} \boldsymbol{\Sigma}_{(11)} & \boldsymbol{\Sigma}_{(21)} \\ \boldsymbol{\Sigma}_{(21)} & \boldsymbol{\Sigma}_{(22)} \end{pmatrix} \begin{pmatrix} \mathbf{I}'_q \\ \mathbf{O}' \end{pmatrix}\end{aligned}$$

Thus $\mathbf{Y} = \mathbf{X}^{(1)} \sim N(\boldsymbol{\mu}^{(1)}, \boldsymbol{\Sigma}_{11})$. Similarly $\mathbf{Y} = \mathbf{X}^{(2)} \sim N(\boldsymbol{\mu}^{(2)}, \boldsymbol{\Sigma}_{22})$. ■

Theorem 5 If $\mathbf{Y}_{(n \times 1)} \sim N(\mathbf{O}, \mathbf{I}_n)$, then $\mathbf{Y}'\mathbf{A}\mathbf{Y}$ is distributed as χ_k^2 , where $\mathbf{A}$ is idempotent of rank k .

Proof. There exists an orthogonal matrix $\mathbf{P}$, such that $\mathbf{P}'\mathbf{A}\mathbf{P} = \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$.

Let $\mathbf{Z} = \mathbf{P}'\mathbf{Y}$. Then $\mathbf{Z} \sim N(\mathbf{P}'\mathbf{O}, \mathbf{P}'\mathbf{I}\mathbf{P})$. But $\mathbf{P}'\mathbf{I}\mathbf{P} = \mathbf{P}'\mathbf{P} = \mathbf{I}$ because $\mathbf{P}$ is orthogonal, i.e. $\mathbf{Z} \sim N(\mathbf{O}, \mathbf{I})$. Thus

$$\begin{aligned}\mathbf{Y}'\mathbf{A}\mathbf{Y} &= \mathbf{Z}'\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{Z} \\ &= \mathbf{Z}' \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} \mathbf{Z} \\ &= (\mathbf{Z}'_1, \mathbf{Z}'_2) \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} (\mathbf{Z}_1, \mathbf{Z}_2) \\ &= \mathbf{Z}'_1 \mathbf{I}_k \mathbf{Z}_1 \\ &= \sum_{i=1}^k Z_i^2\end{aligned}$$

but $\mathbf{Z} \sim N(\mathbf{O}, \mathbf{I}) \implies$ all Z_i are independent. Thus $\mathbf{Y}'\mathbf{A}\mathbf{Y} = \mathbf{Z}'_1 \mathbf{Z}_1 = \sum_{i=1}^k Z_i^2 \sim \chi_k^2$. ■

Theorem 6 If $\mathbf{Y}_{(n \times 1)} \sim N(\mathbf{O}, \mathbf{I}_n)$, then the linear form $\mathbf{B}\mathbf{Y}$ is independent of the quadratic form $\mathbf{Y}'\mathbf{A}\mathbf{Y}$ ($\mathbf{A}$ idempotent of rank k) if $\mathbf{B}\mathbf{A} = \mathbf{O}$.

Proof. From the above theorem, there exists an orthogonal $\mathbf{P}$ such that

$\mathbf{P}'\mathbf{A}\mathbf{P} = \begin{pmatrix} I_k & O \\ O & O \end{pmatrix}$. If $\mathbf{Z} = \mathbf{P}'\mathbf{Y}$, then $\mathbf{Z} \sim \mathbf{N}(\mathbf{O}, \mathbf{I})$ and

$$\mathbf{Y}'\mathbf{A}\mathbf{Y} = \mathbf{Z}'_1\mathbf{Z}_1 = \sum_{i=1}^k \mathbf{z}_i^2 = \mathbf{f}(\mathbf{Z}_1, \dots, \mathbf{Z}_k)$$

Now $\mathbf{O} = \mathbf{B}\mathbf{A} = \mathbf{B}\mathbf{A}\mathbf{P} = \mathbf{B}\mathbf{P}\mathbf{P}'\mathbf{A}\mathbf{P}$

Let $\mathbf{C} = \mathbf{B}\mathbf{P}$, $\mathbf{C}\mathbf{P}'\mathbf{A}\mathbf{P} = \mathbf{O}$, but $\mathbf{P}'\mathbf{A}\mathbf{P} = \begin{pmatrix} I_k & O \\ O & O \end{pmatrix}$

Therefore

$$\begin{aligned} \mathbf{C} \begin{pmatrix} I_k & O \\ O & O \end{pmatrix} &= \mathbf{O} \\ \begin{pmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{pmatrix} \begin{pmatrix} I_k & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} &= \begin{pmatrix} \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} \end{aligned}$$

Thus $\mathbf{C}_{11} = \mathbf{O}$ and $\mathbf{C}_{21} = \mathbf{O}$, or $\mathbf{C}$ is of the form $\mathbf{C} = (\mathbf{O}, \mathbf{C}_2)$.

Thus $\mathbf{B}\mathbf{Y} = \mathbf{B}\mathbf{P}\mathbf{P}'\mathbf{Y} = \mathbf{C}\mathbf{Z} = (\mathbf{O}, \mathbf{C}_2) \begin{pmatrix} \mathbf{Z}_1 \\ \mathbf{Z}_2 \end{pmatrix} = \mathbf{C}_2\mathbf{Z}_2 = \mathbf{g}(\mathbf{Z}_{k+1}, \dots, \mathbf{Z}_n)$ i.e. a function of the remaining elements of $\mathbf{Z}$.

Thus $\mathbf{Y}'\mathbf{A}\mathbf{Y} = \mathbf{f}(\mathbf{Z}_1, \dots, \mathbf{Z}_k)$ and $\mathbf{B}\mathbf{Y} = \mathbf{g}(\mathbf{Z}_{k+1}, \dots, \mathbf{Z}_n)$. But $\mathbf{Z} \sim \mathbf{N}(\mathbf{O}, \mathbf{I})$ indicating that all the elements of $\mathbf{Z}$ are independent. This implies that $\mathbf{Y}'\mathbf{A}\mathbf{Y}$ and $\mathbf{B}\mathbf{Y}$ are independent. ■

Theorem 7 The maximum likelihood estimate $\hat{\beta}$ is distributed $N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$.

Proof. In what follows we will assume $\mathbf{e} \sim \mathbf{N}(\mathbf{0}, \sigma^2\mathbf{I}_n)$. Now

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{B}\mathbf{Y}$$

Since $\mathbf{e} \sim \mathbf{N}(\mathbf{0}, \sigma^2\mathbf{I}_n)$ it follows that $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ is distributed $\mathbf{N}(\mathbf{X}\beta, \sigma^2\mathbf{I}_n)$ so that the linear combination $\mathbf{B}\mathbf{Y}$ is distributed $\mathbf{N}(\mathbf{B}\mathbf{X}\beta, \sigma^2\mathbf{B}\mathbf{B}')$. Now

$$\mathbf{B}\mathbf{X}\beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta = \mathbf{I}_k\beta = \beta$$

The covariance matrix of $\hat{\beta}$ is

$$\begin{aligned} \text{cov}(\hat{\beta}) &= \sigma^2\mathbf{B}\mathbf{B}' \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X})' \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

since $(\mathbf{X}'\mathbf{X})$ and $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric. ■

Theorem 8 *The MLE $\hat{\sigma}^2$ is a biased estimate of σ^2 .*

Proof.

$$\begin{aligned} n\hat{\sigma}^2 &= (\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) \\ &= (\mathbf{Y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y})'(\mathbf{Y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}) \\ &= \{(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{Y}\}'\{(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{Y}\} \\ &= \mathbf{Y}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{Y} \end{aligned}$$

since $\mathbf{A} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{A}'$ being symmetric. Thus $n\hat{\sigma}^2 = \mathbf{Y}'\mathbf{A}^2\mathbf{Y}$

But

$$\begin{aligned} \mathbf{A}^2 &= (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \\ &\quad \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ &= \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{A} \end{aligned}$$

hence $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{A}' = \mathbf{A}$ indicating that $\mathbf{A}$ is idempotent. Therefore

$$n\hat{\sigma}^2 = \mathbf{Y}'\mathbf{A}\mathbf{Y} = (\mathbf{X}\beta + \mathbf{e})'\mathbf{A}(\mathbf{X}\beta + \mathbf{e})$$

But

$$\mathbf{A}\mathbf{X} = (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{X} = \mathbf{X} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$$

Similarly $\mathbf{X}'\mathbf{A} = \mathbf{0}$ with

$$n\hat{\sigma}^2 = \mathbf{e}'\mathbf{A}\mathbf{e} \tag{33}$$

Taking expected values

$$\begin{aligned} E(\hat{\sigma}^2) &= \frac{1}{n}E(\mathbf{e}'\mathbf{A}\mathbf{e}) = \frac{1}{n}E\left(\sum_{i=1}^n \sum_{j=1}^n a_{ij}e_ie_j\right) \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n E(e_ie_j) \end{aligned}$$

But $E(e_i^2) = \sigma^2$ and $E(e_i e_j) = 0$ for $i \neq j$. Therefore

$$\begin{aligned}
 E(\hat{\sigma}^2) &= \frac{1}{n} \sigma^2 \sum_{i=1}^n a_{ii} = \frac{\sigma^2}{n} \text{tr}(\mathbf{A}) = \frac{\sigma^2}{n} \text{tr}(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \\
 &= \frac{\sigma^2}{n} (\text{tr}\mathbf{I}_n - \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')) \\
 &= \frac{\sigma^2}{n} (n - \text{tr}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) \text{ using } \text{tr}(\mathbf{CD}) = \text{tr}(\mathbf{DC}) \\
 &= \frac{\sigma^2}{n} (n - k) \text{ since } \mathbf{X}'\mathbf{X} \text{ is a } (k \times k) \text{ matrix.}
 \end{aligned}$$

Thus $E(\hat{\sigma}^2) \neq \sigma^2$ and $\hat{\sigma}^2$ is a biased estimate of σ^2 . The theorem is proved.

In the proof we have not used the fact that $\mathbf{e}$ is distributed $N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ but only that $E(\mathbf{e}) = \mathbf{0}$ and $E(\mathbf{e}\mathbf{e}') = \sigma^2 \mathbf{I}_n$. ■

Theorem 9 In the model $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ where $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$

$\frac{n-k}{\sigma^2} s^2$ is distributed as a χ^2 variate with $n-k$ degrees of freedom

Proof. From equation (33) it follows that

$$(n-k)s^2 = n\hat{\sigma}^2 = (\mathbf{Y} - \mathbf{X}\hat{\beta})(\mathbf{Y} - \mathbf{X}\hat{\beta})' = \mathbf{Y}'\mathbf{A}\mathbf{Y} = \mathbf{e}'\mathbf{A}\mathbf{e}$$

It has been shown that $\mathbf{A}$ is idempotent. Assume the rank of $\mathbf{A}$ is f . Then

$$\frac{\mathbf{e}}{\sigma} \sim N(\mathbf{0}, \mathbf{I}_n) \text{ and } \frac{n-k}{\sigma^2} s^2 = \frac{\mathbf{e}'}{\sigma} \mathbf{A} \frac{\mathbf{e}}{\sigma}$$

is distributed as χ_f^2 since $\mathbf{A}$ is idempotent of rank f . Now we need to calculate f . Since $\mathbf{A}$ is idempotent, there exist an orthogonal matrix $\mathbf{P}$ such that

$$\mathbf{P}'\mathbf{A}\mathbf{P} = \begin{pmatrix} \mathbf{I}_f & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$$

Since $\mathbf{P}\mathbf{P}' = \mathbf{I}$

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}\mathbf{P}\mathbf{P}') = \text{tr}(\mathbf{P}'\mathbf{A}\mathbf{P}) = \text{tr} \begin{pmatrix} \mathbf{I}_f & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} = f$$

Hence the degrees of freedom $f = \text{tr}(\mathbf{A}) = \text{tr}(\mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = n - k$. ■

Theorem 10 If $\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$ with $\mathbf{e} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ then

$$\hat{\beta} \text{ and } \frac{n-k}{\sigma^2} s^2$$

are independently distributed.

Proof.

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$$

which is a linear form in $\mathbf{Y}$. Also

$$(n-k)s^2 = \mathbf{Y}'\mathbf{A}\mathbf{Y}$$

which is a quadratic form in $\mathbf{Y}$. But

$$\begin{aligned} \mathbf{B}\mathbf{A} &= (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}') \\ &= (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}' - \mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}') \\ &= (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}' - \mathbf{X}') \\ &= \mathbf{0} \end{aligned}$$

The result follows from Theorem 5. ■

Theorem 11 If $\mathbf{Y}$ ($q \times 1$) has a multivariate normal distribution $N(\mathbf{0}, \Sigma)$ then $\mathbf{Y}'\Sigma^{-1}\mathbf{Y}$ is distributed χ_q^2 .

Proof. There exist a non-singular matrix $\mathbf{C}$ ($q \times q$) such that $\mathbf{C}'\Sigma^{-1}\mathbf{C} = \mathbf{I}$. Let $\mathbf{Y} = \mathbf{C}\mathbf{Z}$ or $\mathbf{Z} = \mathbf{C}^{-1}\mathbf{Y}$. Then

$$\mathbf{Y}'\Sigma^{-1}\mathbf{Y} = \mathbf{Z}'\mathbf{C}'\Sigma^{-1}\mathbf{C}\mathbf{Z} = \mathbf{Z}'\mathbf{I}\mathbf{Z} = \mathbf{Z}'\mathbf{Z} = \sum_{i=1}^q Z_i^2.$$

But $\mathbf{Z} = \mathbf{C}^{-1}\mathbf{Y}$ is distributed $N(\mathbf{0}, \mathbf{C}^{-1}\Sigma(\mathbf{C}^{-1})')$ being a linear combination of $\mathbf{Y}$. But $\mathbf{C}'\Sigma^{-1}\mathbf{C} = \mathbf{I}$, so that $\mathbf{C}^{-1}\Sigma(\mathbf{C}^{-1})' = (\mathbf{C}'\Sigma^{-1}\mathbf{C})^{-1} = \mathbf{I}^{-1} = \mathbf{I}$. Also $\mathbf{C}^{-1}\mathbf{C} = \mathbf{I}$ or $(\mathbf{C}^{-1}\mathbf{C})' = \mathbf{C}'(\mathbf{C}^{-1})' = \mathbf{I}' = \mathbf{I}$ so that $(\mathbf{C}^{-1})' = (\mathbf{C}')^{-1}$ and hence $\mathbf{C}^{-1}\Sigma(\mathbf{C}^{-1})' = \mathbf{C}^{-1}\Sigma(\mathbf{C}')^{-1} = \mathbf{I}$. Thus $\mathbf{Z} \sim N(\mathbf{0}, \mathbf{I})$ and $\mathbf{Z}'\mathbf{Z} \sim \chi_q^2$. ■

Proposition 12 $s^2 = \frac{1}{n-k}(\mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y})$

Proof. Since

$$(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = \mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\hat{\beta} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta}$$

but $(\mathbf{Y}'\mathbf{X}\hat{\beta})' = \hat{\beta}'\mathbf{X}'\mathbf{Y}$ being a scalar, and $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{Y}$ from the normal equations. Hence

$$(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = \mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y}$$

and

$$s^2 = \frac{1}{n-k}(\mathbf{Y}'\mathbf{Y} - \hat{\beta}'\mathbf{X}'\mathbf{Y})$$

■

13 Useful References

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